

Visualization Design:

Effectiveness Principles

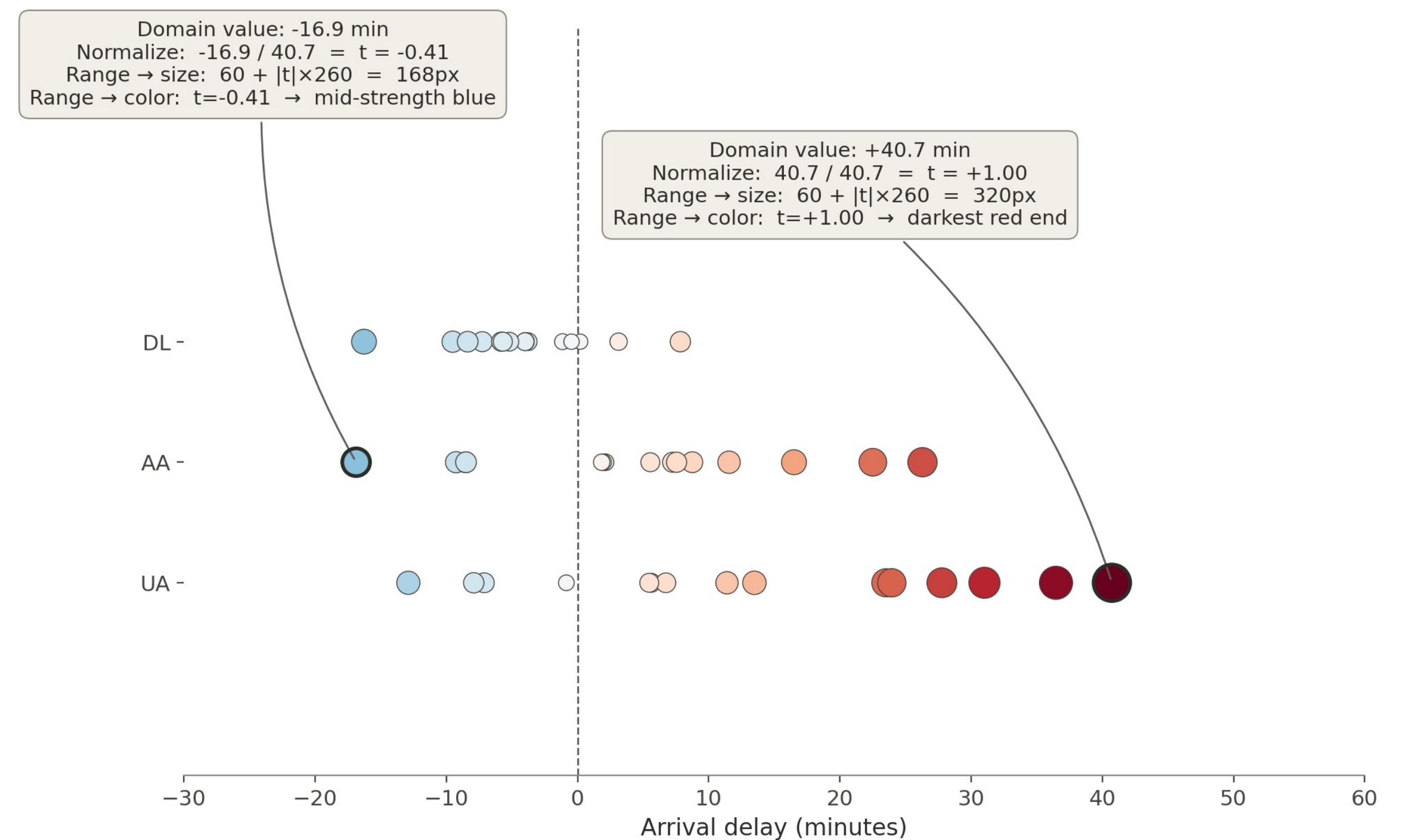
Sarah Groves

What you will learn today

- Guidelines on **scale**: why does range matter?
- **Expressiveness** and **effectiveness** principles
- Channel **ranking**

Dimensions of a scale

- **Domain:** the data values to be mapped
- **Range:** the visual channel values derived from domain
- **Type:** sequential, diverging, linear, log, power, square root, ...
- **Aggregator:** how values are aggregated if a mark represents multiple items



Guidelines: Scales for Position/Length Encodings

I know the **domain** of my attribute.

How do I decide what the **range** of my channel encoding should be?

Design Considerations

Starting from Zero

Aspect ratio: deciding on range

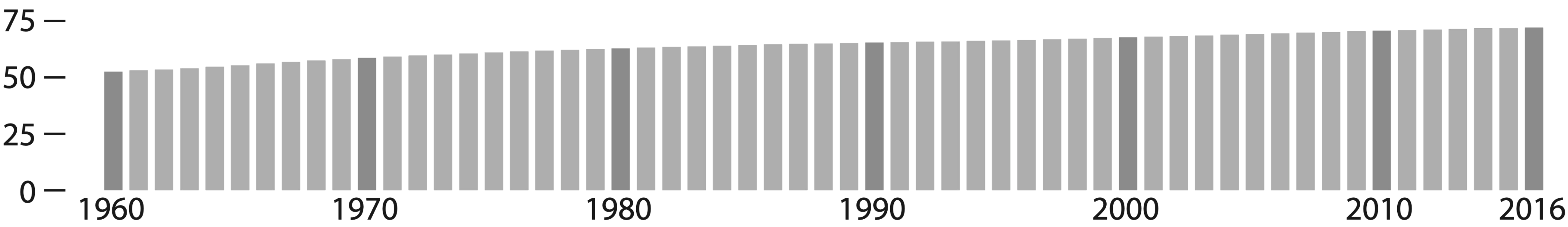
Linear vs. Log Scale

Outliers and Scale Break

All Charts Should Start at Zero

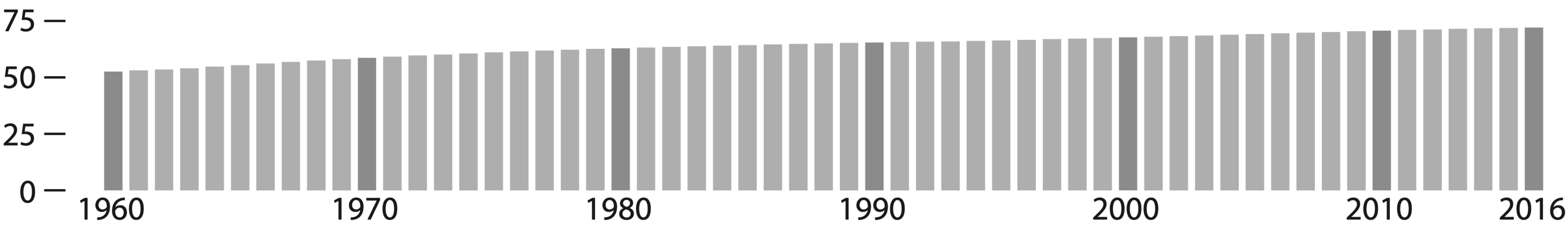
Myth or Truth?

Average world life expectancy at birth (years)



(Source: World Bank Data)

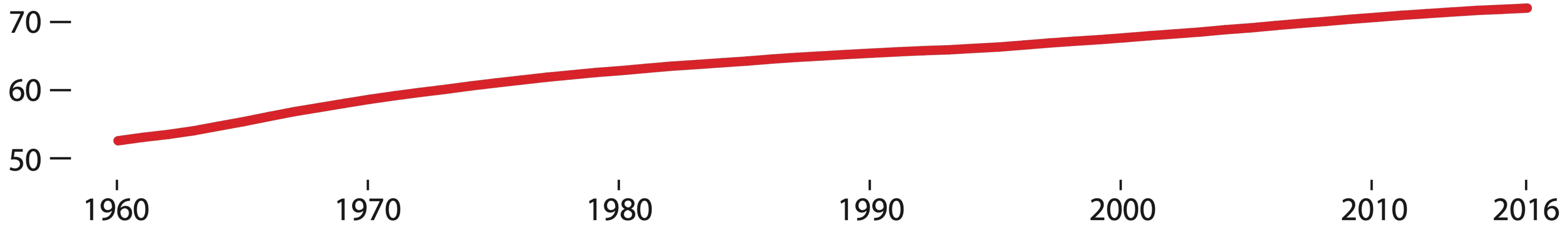
Average world life expectancy at birth (years)

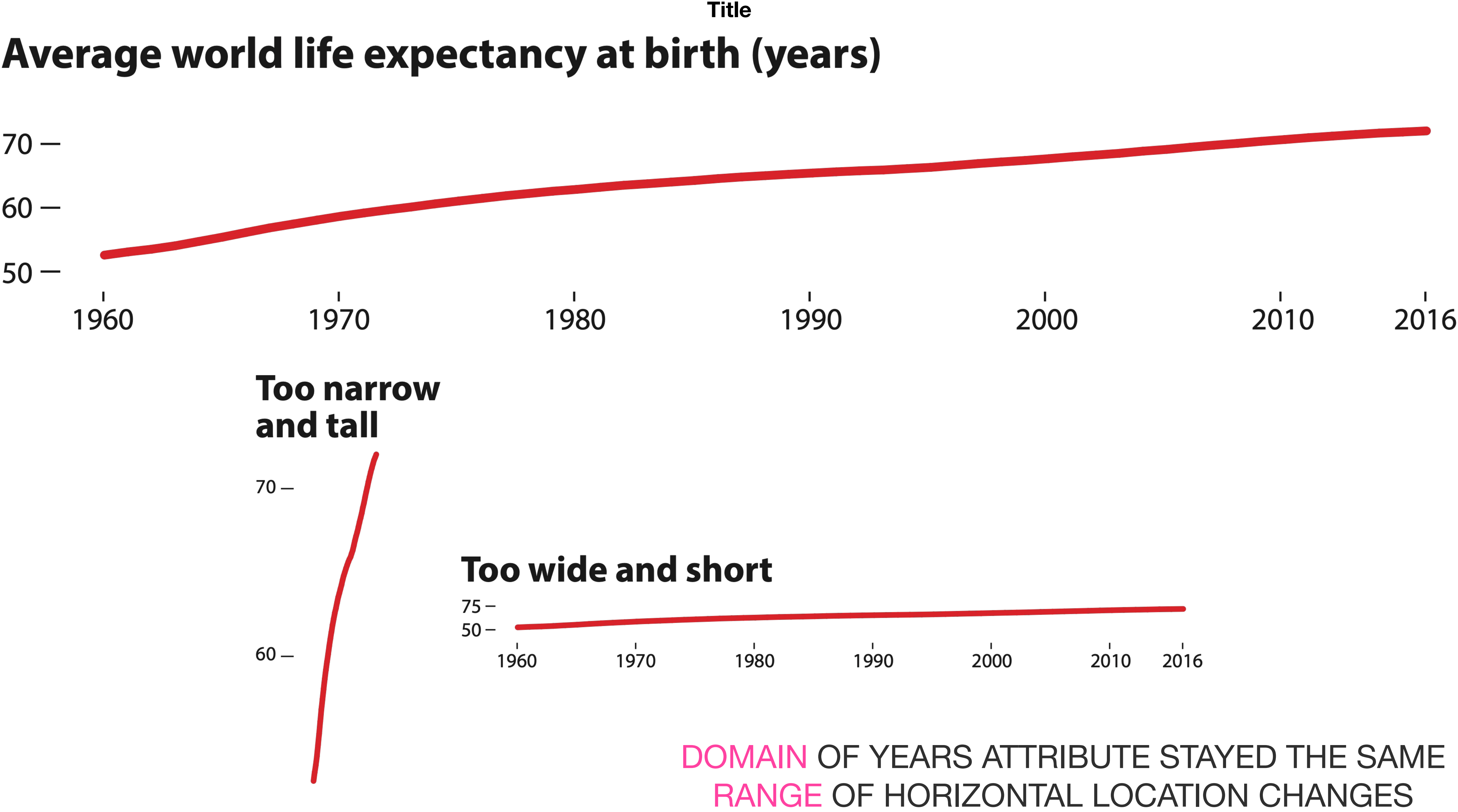


(Source: World Bank Data)

Title

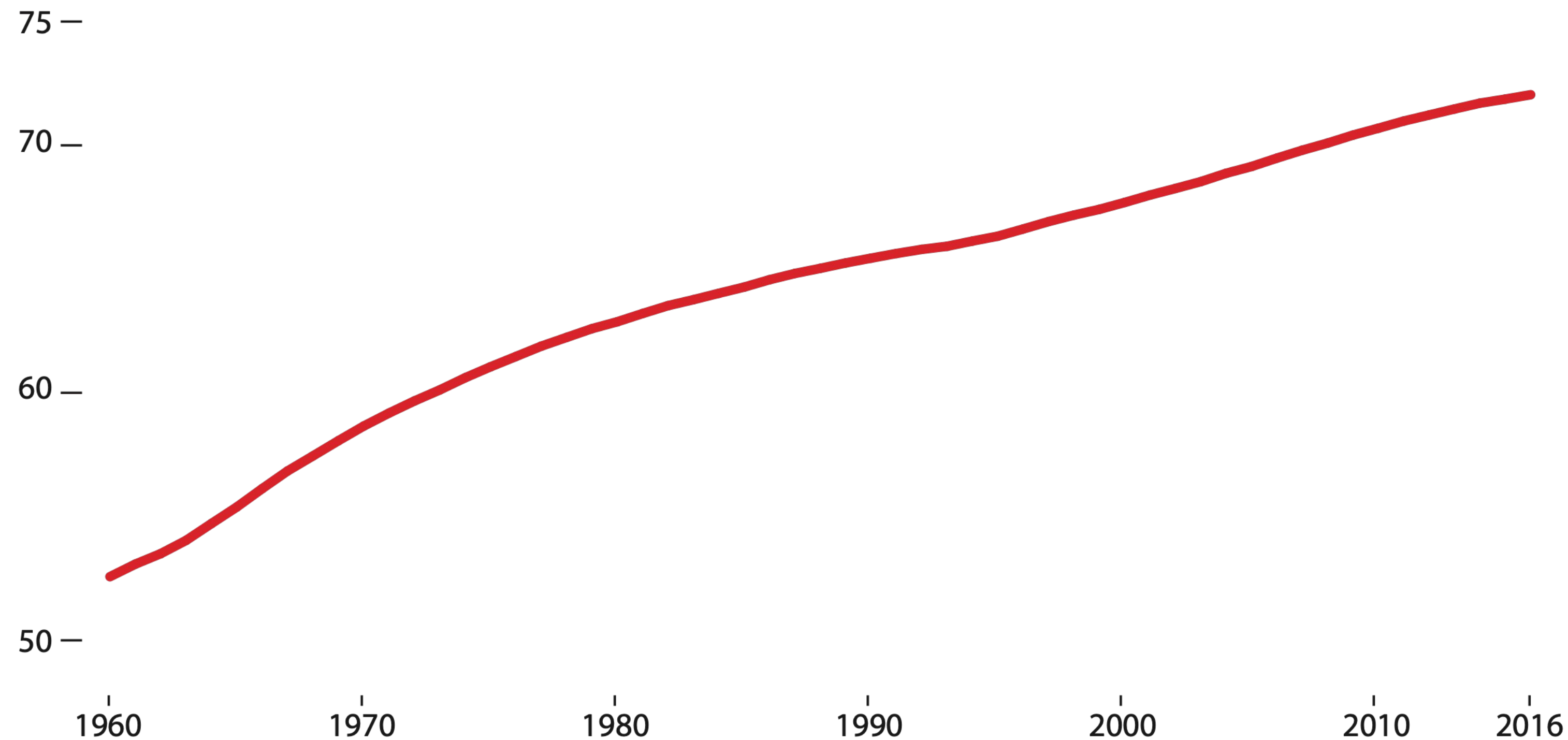
Average world life expectancy at birth (years)





Aspect Ratio

Average world life expectancy at birth (years)



Approximate the proportion of the chart to match the depicted trend

35% increase $\approx 1/3_{\text{rd}}$
 $\approx 3:1$ aspect ratio

Guideline if relevant data are far from zero

- In the life expectancy plots we just looked at, we **avoided the “start at 0”** decision by switching from a bar plot to a line plot
- Because we had a **line plot**, it became more important to ensure the **aspect ratio** was proportional to the true increase in life expectancy than keeping 0 on the plot
- If you don't start at 0: **Center the y-axis on the data's mean**, then extend the y-axis 0.75 SD above and below the mean (Witt 2019)

Guidelines: Scales for Position/Length Encodings

Design Considerations

Starting from Zero

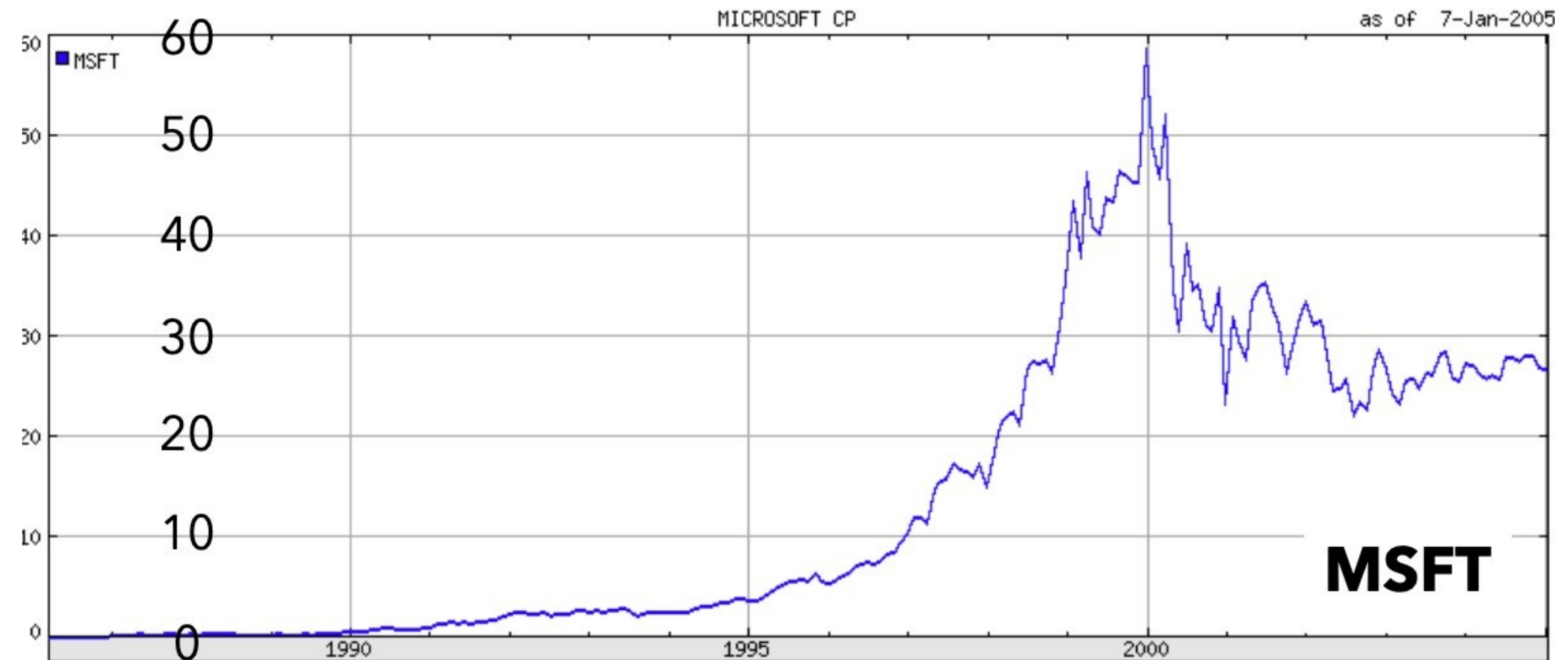
Aspect ratio: deciding on range

Linear vs. Log Scale

Outliers and Scale Break

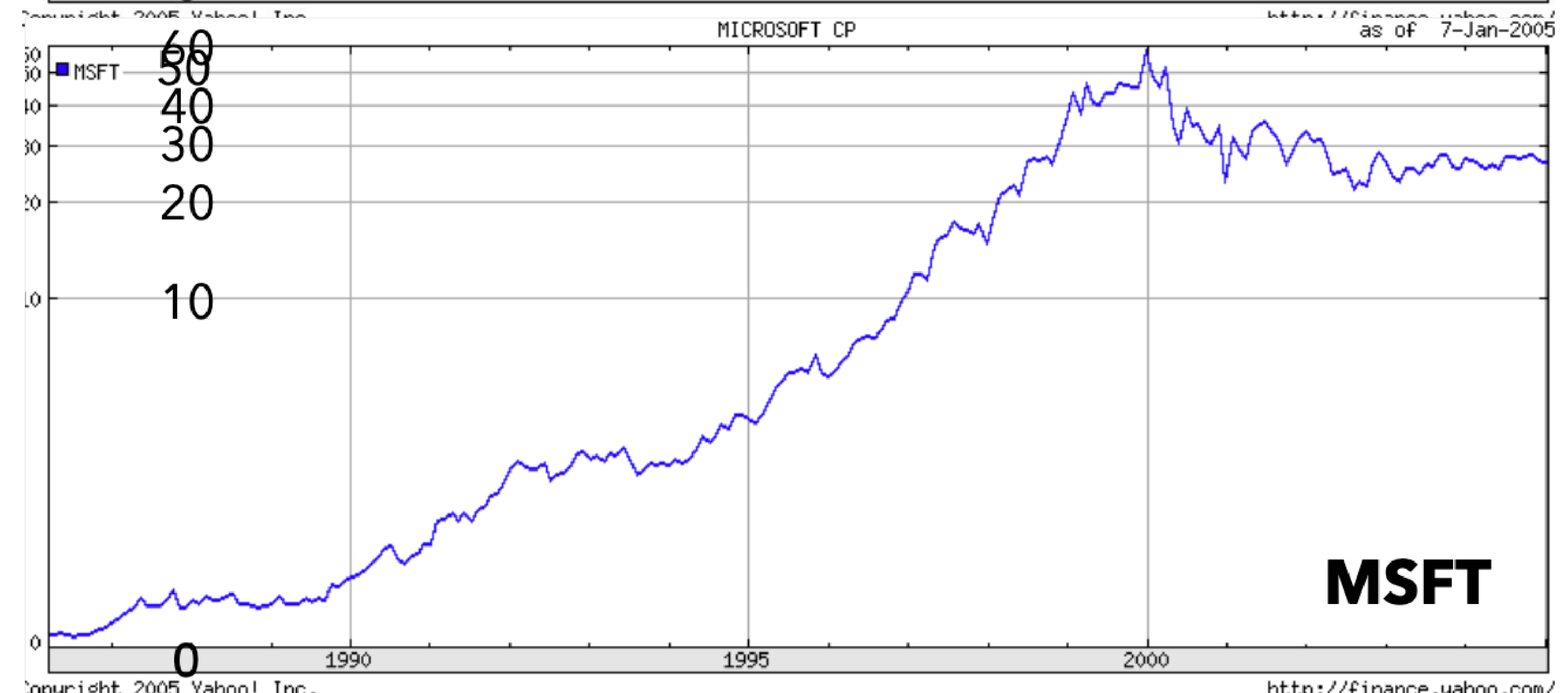
Scale type: linear, log, power, square root...

Linear Scale
Absolute Change



Log Scale
Small fluctuations
Percent change

DOMAIN IS THE SAME
RANGE IS THE SAME
SCALE TYPE HAS CHANGED
=



Same data is mapped to different vertical locations

When to apply a log scale?

Address data skew (e.g., long tails, outliers)

Enables comparison within and across multiple orders of magnitude.

Focus on multiplicative factors (not additive) Recall that the logarithm transforms $\times$ to $+$! Percentage change, not absolute value.

Constraints

positive, non-zero values
audience familiarity

Linear scales show absolute change, but **log scales** are better when there are small fluctuations or percentage change.

Guidelines: Scales for Position/Length Encodings

Design Considerations

Starting from Zero

Aspect ratio: deciding on range

Linear vs. Log Scale

Outliers and Scale Break

userratio

timetaken

150

100

50

0

10

15

25

30

35

38

45

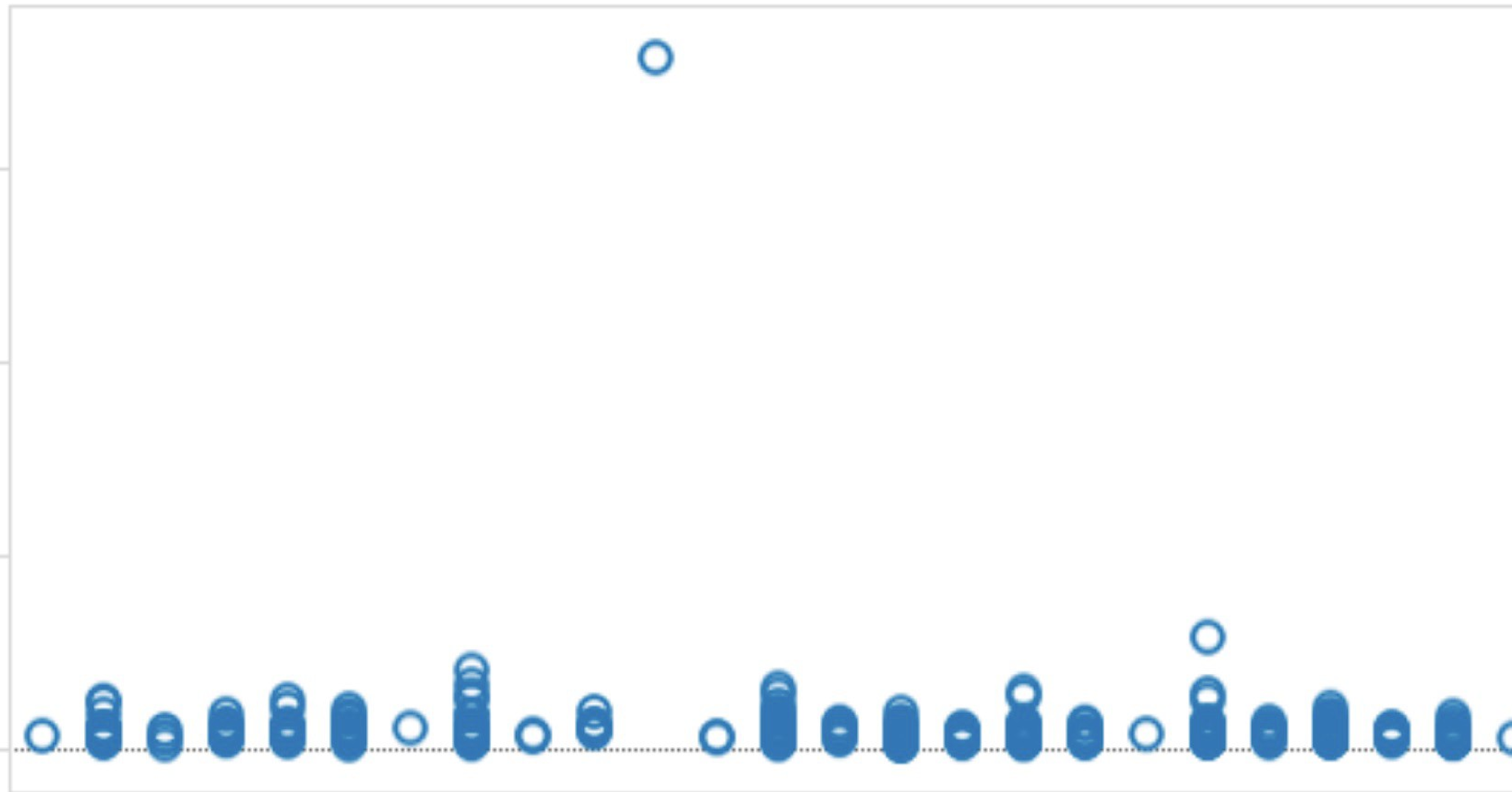
55

65

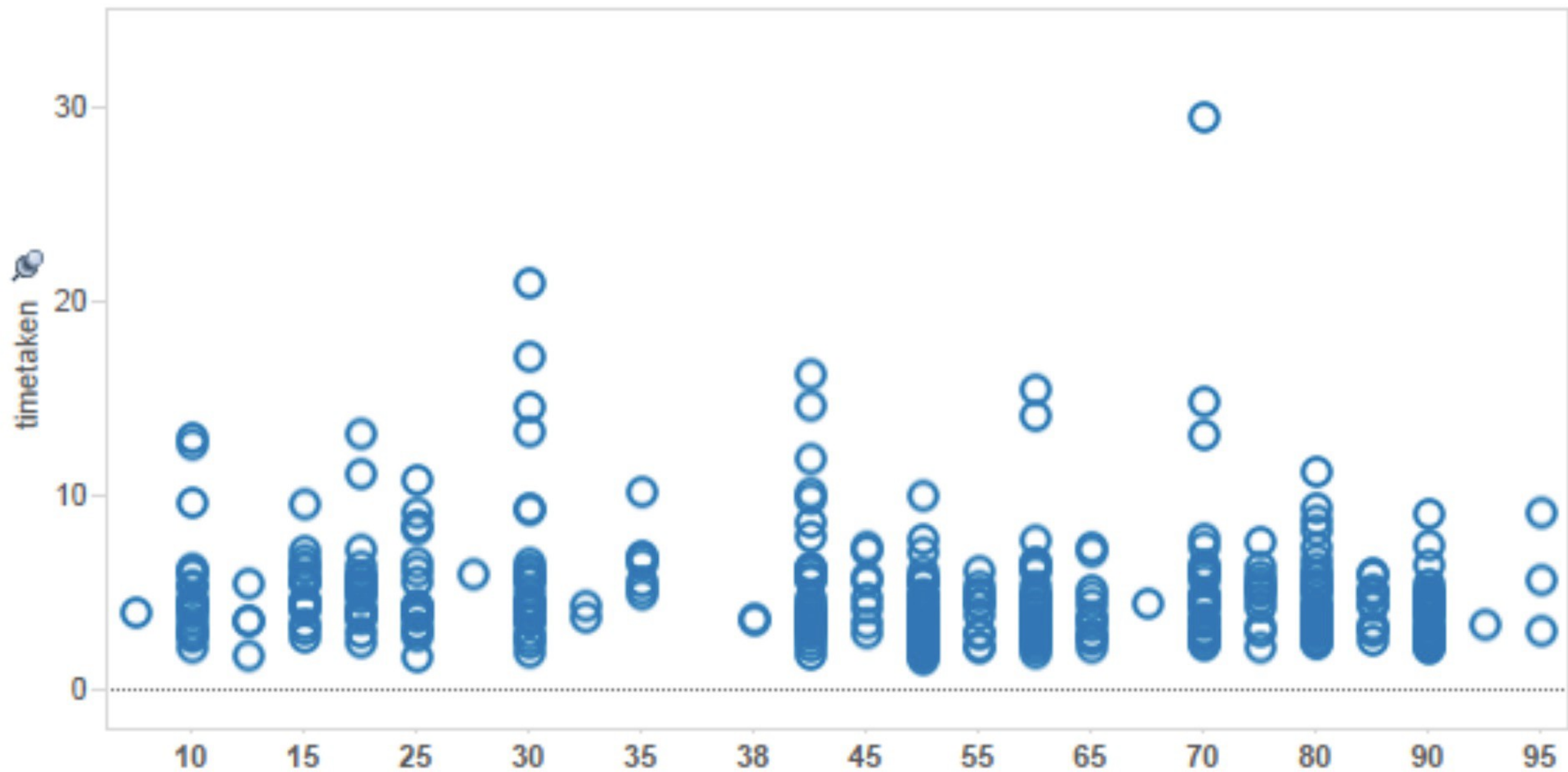
70

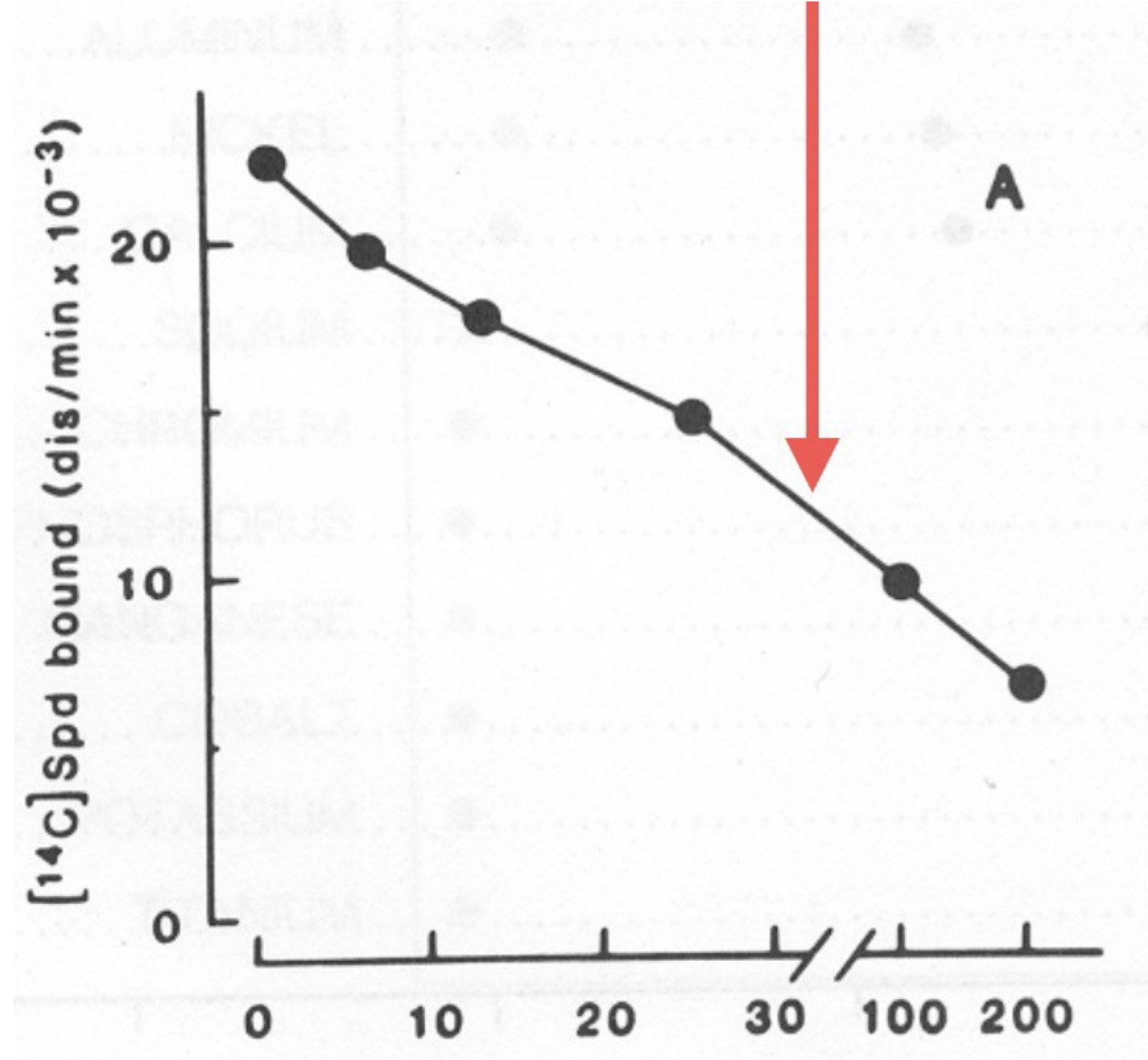
80

90

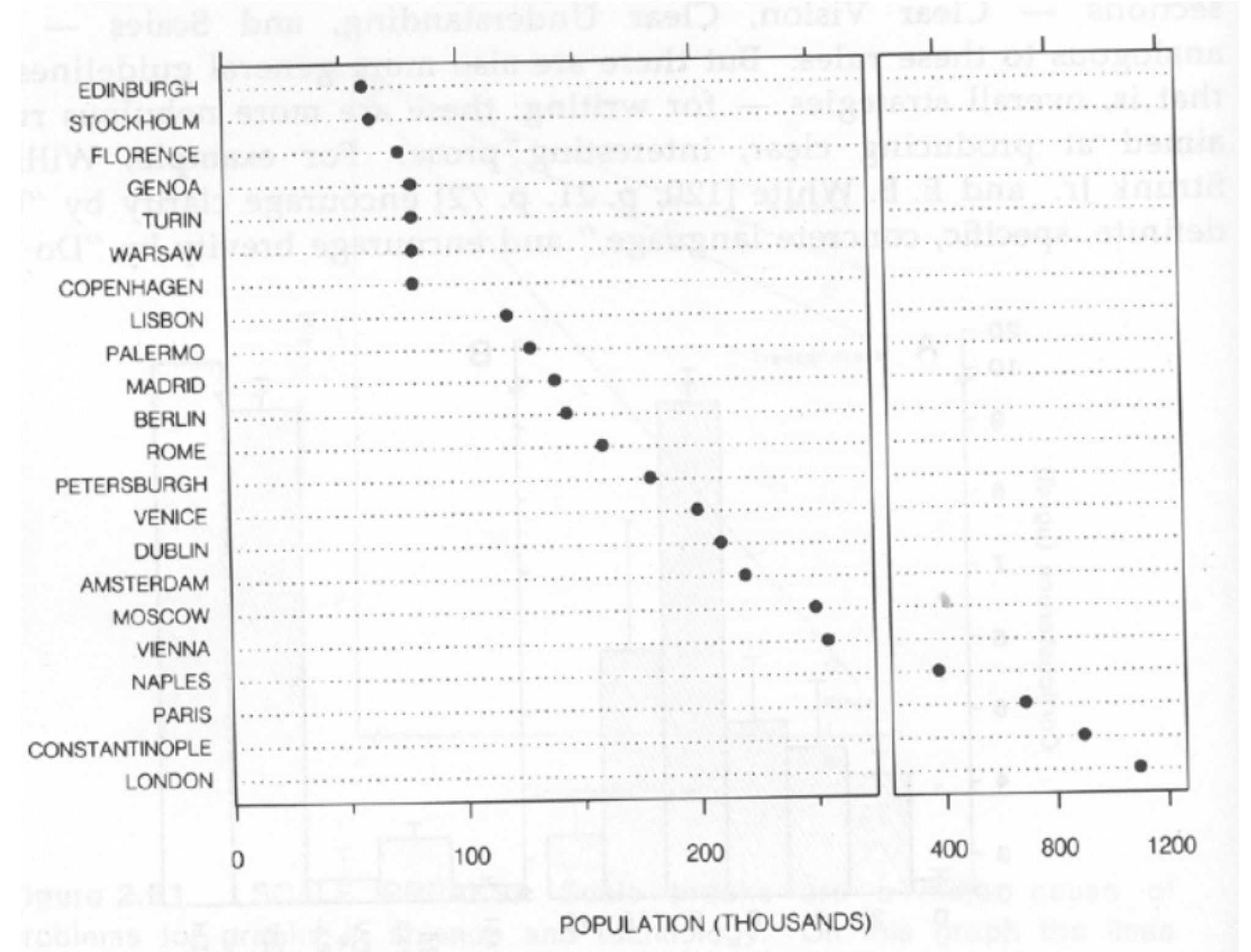


userratio

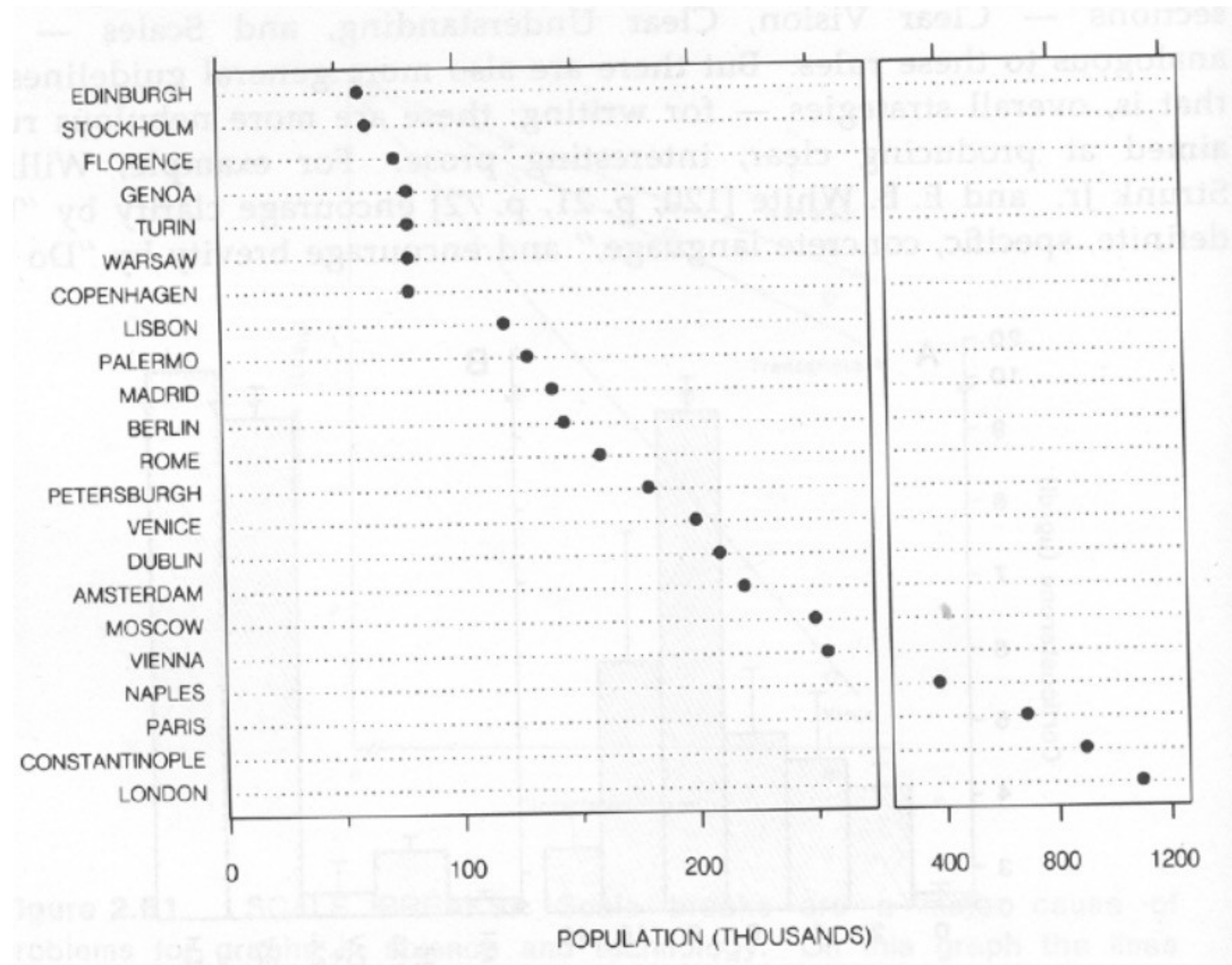




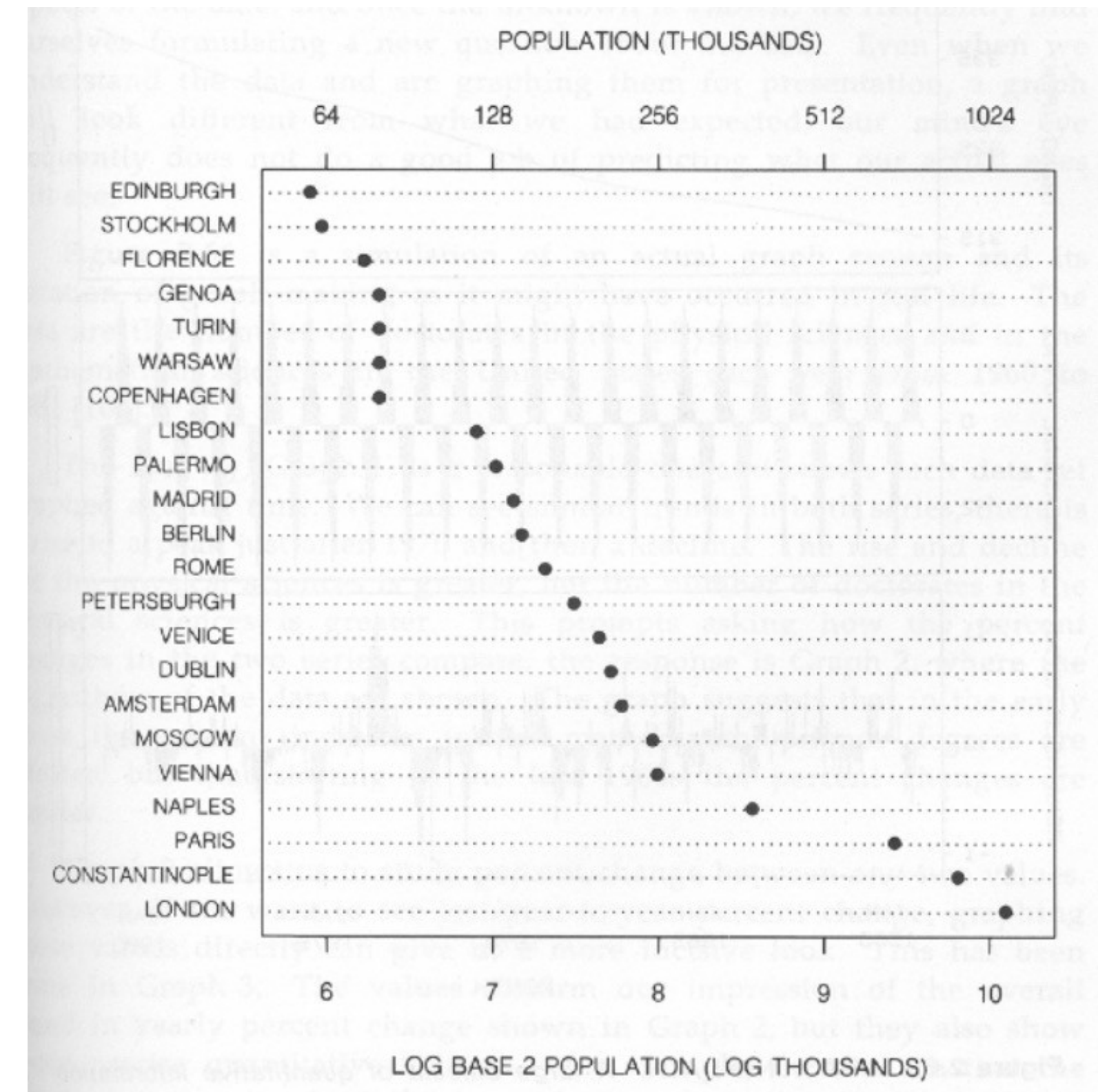
Poor scale break



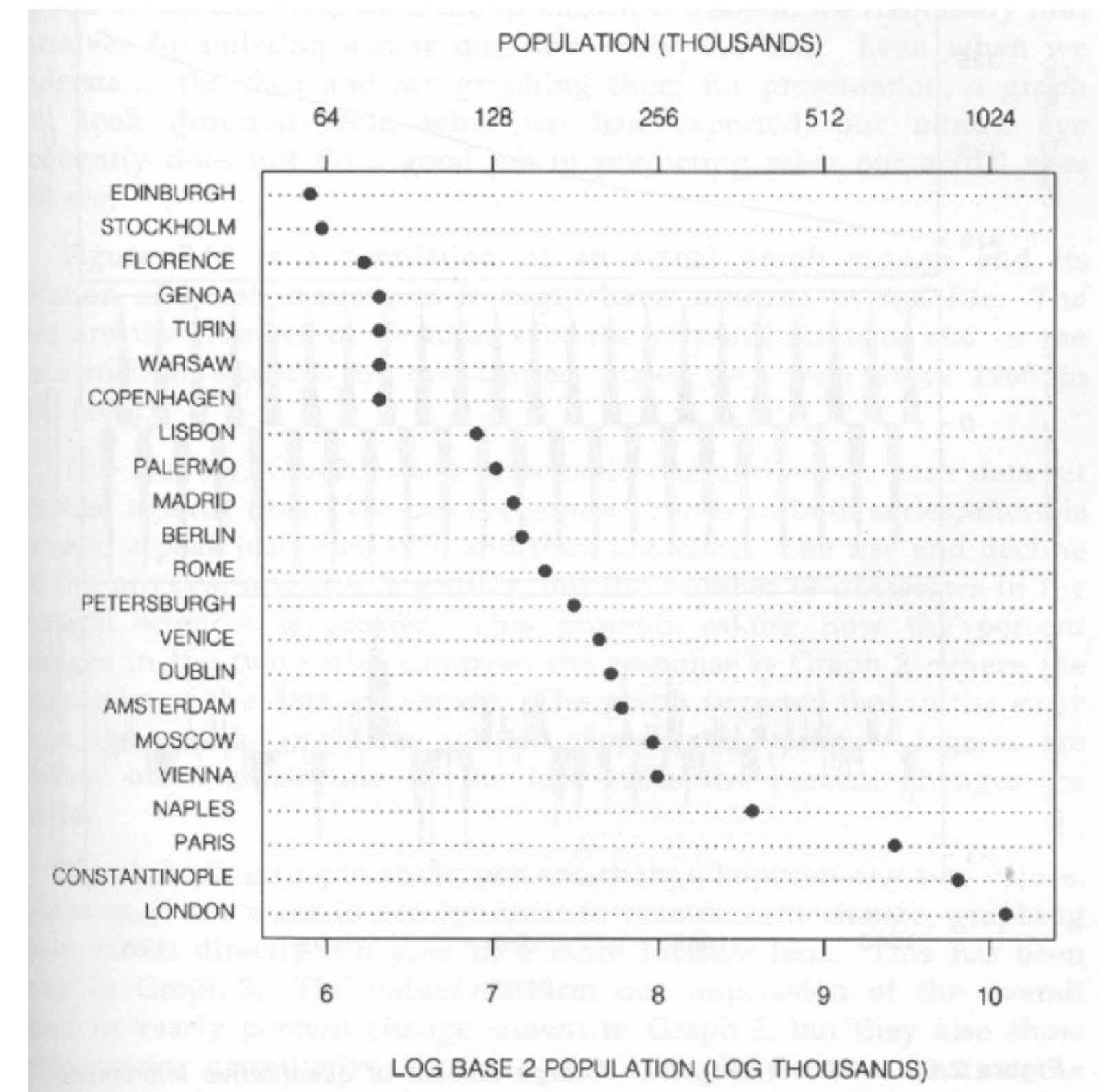
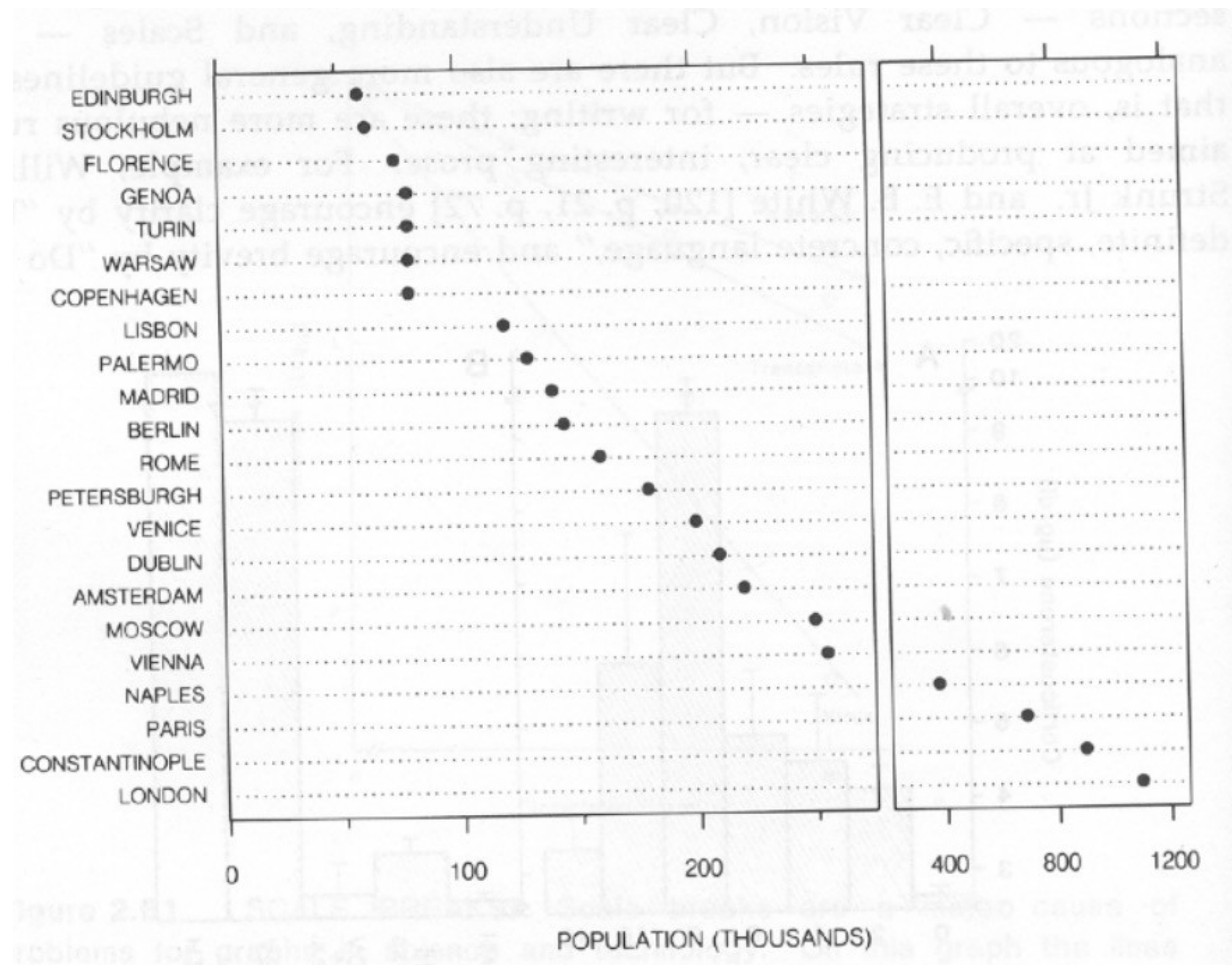
Well-marked scale break



Scale break

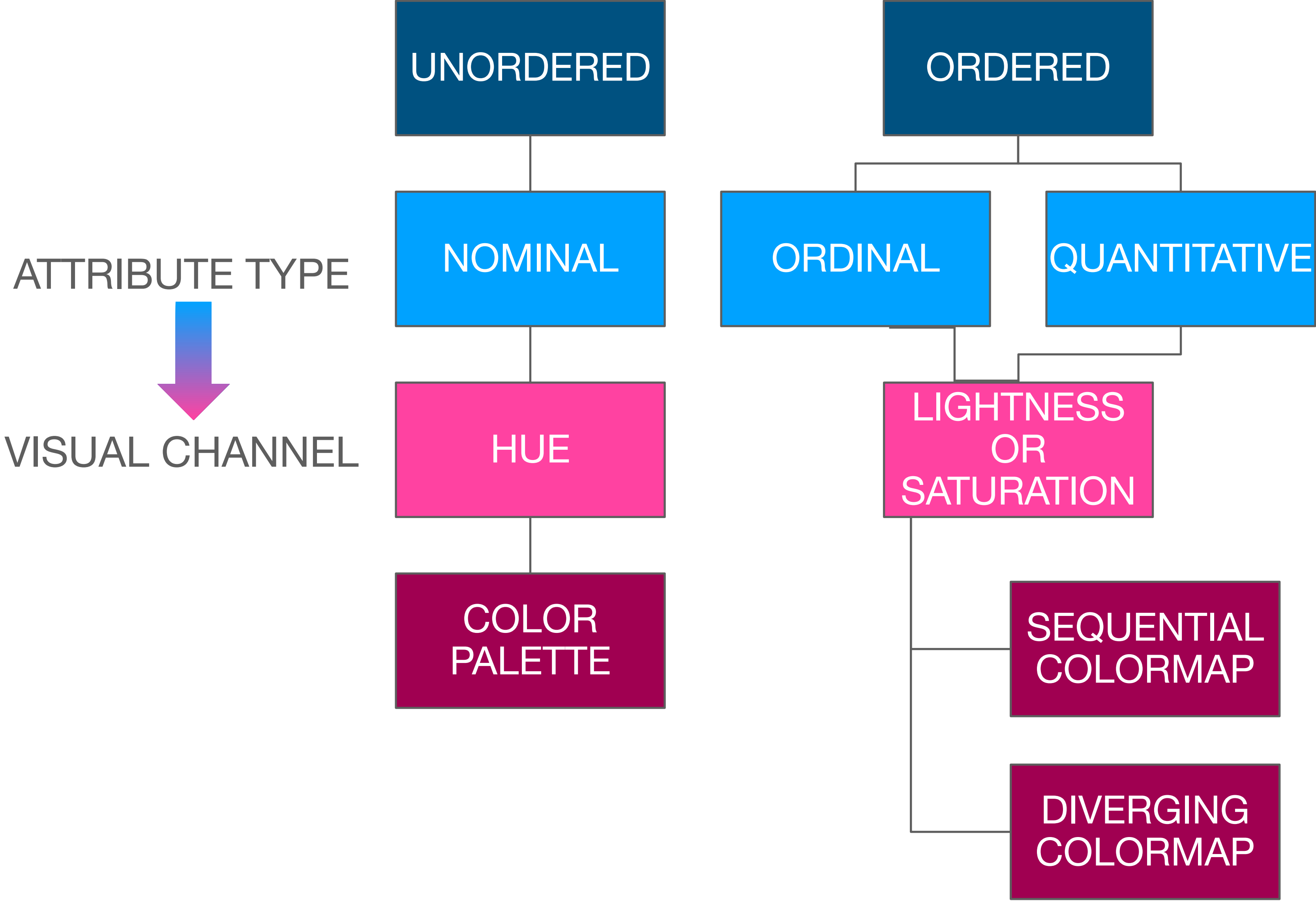


Log Scale



- *Scale break: difficult to compare (cognitive - not perceptual - work)*
 - *Log scale: direct comparison of all data*

Guidelines: Color Scales for Quantitative Data



Sequential color scale

You can still use hues in addition to value/lightness

`# d3.interpolateViridis(t) <>`



Given a number t in the range $[0,1]$, returns the corresponding color from the “viridis” perceptually-uniform color scheme designed by [van der Walt, Smith and Firing](#) for matplotlib, represented as an RGB string.

`# d3.interpolateInferno(t) <>`



Given a number t in the range $[0,1]$, returns the corresponding color from the “inferno” perceptually-uniform color scheme designed by [van der Walt and Smith](#) for matplotlib, represented as an RGB string.

`# d3.interpolateMagma(t) <>`



Given a number t in the range $[0,1]$, returns the corresponding color from the “magma” perceptually-uniform color scheme designed by [van der Walt and Smith](#) for matplotlib, represented as an RGB string.

`# d3.interpolatePlasma(t) <>`



Given a number t in the range $[0,1]$, returns the corresponding color from the “plasma” perceptually-uniform color scheme designed by [van der Walt and Smith](#) for matplotlib, represented as an RGB string.

`# d3.interpolateCividis(t) <>`



Diverging color scale

Uses both hue and lightness as the channel!

```
# d3.interpolateBrBG(t) <>  
# d3.schemeBrBG[k]
```



Given a number t in the range $[0,1]$, returns the corresponding color from the “BrBG” diverging color scheme represented as an RGB string.

```
# d3.interpolatePRGn(t) <>  
# d3.schemePRGn[k]
```



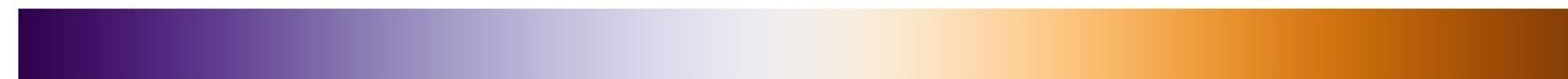
Given a number t in the range $[0,1]$, returns the corresponding color from the “PRGn” diverging color scheme represented as an RGB string.

```
# d3.interpolatePiYG(t) <>  
# d3.schemePiYG[k]
```



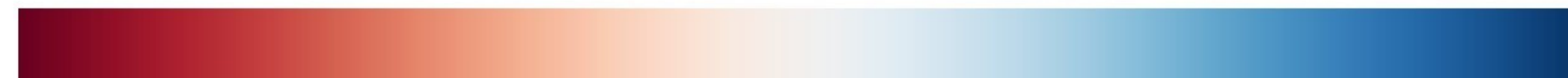
Given a number t in the range $[0,1]$, returns the corresponding color from the “PiYG” diverging color scheme represented as an RGB string.

```
# d3.interpolatePuOr(t) <>  
# d3.schemePuOr[k]
```



Given a number t in the range $[0,1]$, returns the corresponding color from the “PuOr” diverging color scheme represented as an RGB string.

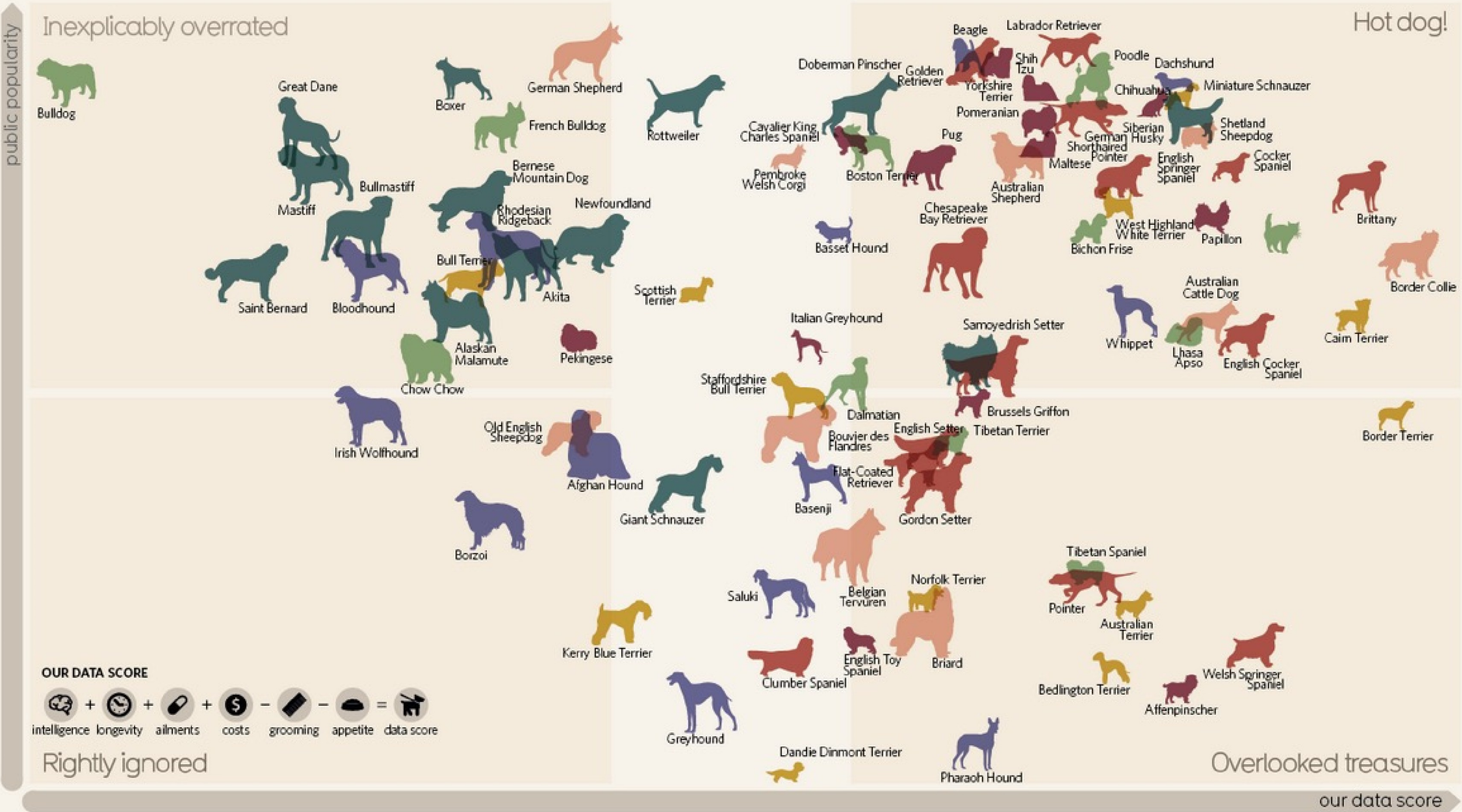
```
# d3.interpolateRdBu(t) <>  
# d3.schemeRdBu[k]
```



**How do we
make good
graphics?**

Best in Show

The ultimate data-dog



What do you notice first?

What takes some time to understand?

Attribute	Data Type
Variety	N
Group	N
Size	O
Smartness	N
Popularity	Q
Ranking	Q

Design Space of Attribute: Channel Mapping

- n data attributes
- k visual channels
- all the possible combinations of encodings (i.e. mappings): $(k+1)^n$
- k = 6 common visual channels: x, y, value, color, size, shape
- n = 3 → 343 possible combinations of encodings
- n = 6 → 117,649 possible combinations of encodings

Many of these encodings won't make sense

Design Criteria

[Mackinlay 1986]

Expressiveness

A set of facts is *expressible* in a visual language if the sentences (i.e. the visualizations) in the language express all the facts in the set of data, and only the facts in the data.

Data

Effectiveness

A visualization is more *effective* than another visualization if the information conveyed by one visualization is more readily perceived than the information in the other visualization.

Visual

Design Criteria *Translated*

[Heer]

Effectiveness:

Use encodings that people decode better
(where better = faster and/or more accurate)

Expressiveness:

Tell the truth and nothing but the truth
(don't lie, and don't lie by omission)

Design Criteria *Translated*

[Heer]

Effectiveness:

Use encodings that people decode better

(where better = faster and/or more accurate)

Expressiveness:

Tell the truth and nothing but the truth

(don't lie, and don't lie by omission)

Effectiveness Rankings

[Mackinlay 1986]

Quantitative

- Position
- Length
- Angle
- Slope
- Area (Size)
- Volume
- Density (Value)
- Color Saturation
- Color Hue
- Texture
- Connection
- Containment
- Shape

Ordinal

- Position
- Density (Value)
- Color Saturation
- Color Hue
- Texture
- Connection
- Containment
- Length
- Angle
- Slope
- Area (Size)
- Volume
- Shape

Nominal

- Position
- Color Hue
- Texture
- Connection
- Containment
- Density (Value)
- Color Saturation
- Shape
- Length
- Angle
- Slope
- Area
- Volume

Effectiveness Rankings

[Mackinlay 1986]

Quantitative	Ordinal	Nominal
Position	Position	Position
Length	Density (Value)	Color Hue
Angle	Color Saturation	Texture
Slope	Color Hue	Connection
Area (Size)	Texture	Containment
Volume	Connection	Density (Value)
Density (Value)	Containment	Color Saturation
Color Saturation	Length	Shape
Color Hue	Angle	Length
Texture	Slope	Angle
Connection	Area (Size)	Slope
Containment	Volume	Area
Shape	Shape	Volume

Effectiveness Rankings

[Mackinlay 1986]

Quantitative

- Position
- Length
- Angle
- Slope
- Area (Size)
- Volume
- Density (Value)
- Color Saturation
- Color Hue**
- Texture
- Connection
- Containment
- Shape

Ordinal

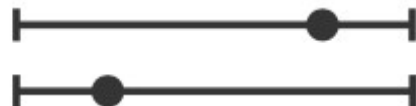
- Position
- Density (Value)
- Color Saturation
- Color Hue**
- Texture
- Connection
- Containment
- Length
- Angle
- Slope
- Area (Size)
- Volume
- Shape

Nominal

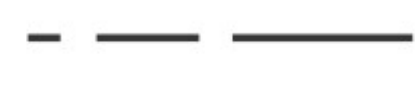
- Position
- Color Hue**
- Texture
- Connection
- Containment
- Density (Value)
- Color Saturation
- Shape
- Length
- Angle
- Slope
- Area
- Volume

Modernization of Visual Channels

➔ **Magnitude Channels:** **O** or **Q** attributes

Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

Same

Most
Effectiveness
Least

➔ **Identity Channels:** **N** attributes

Spatial region 

Color hue 

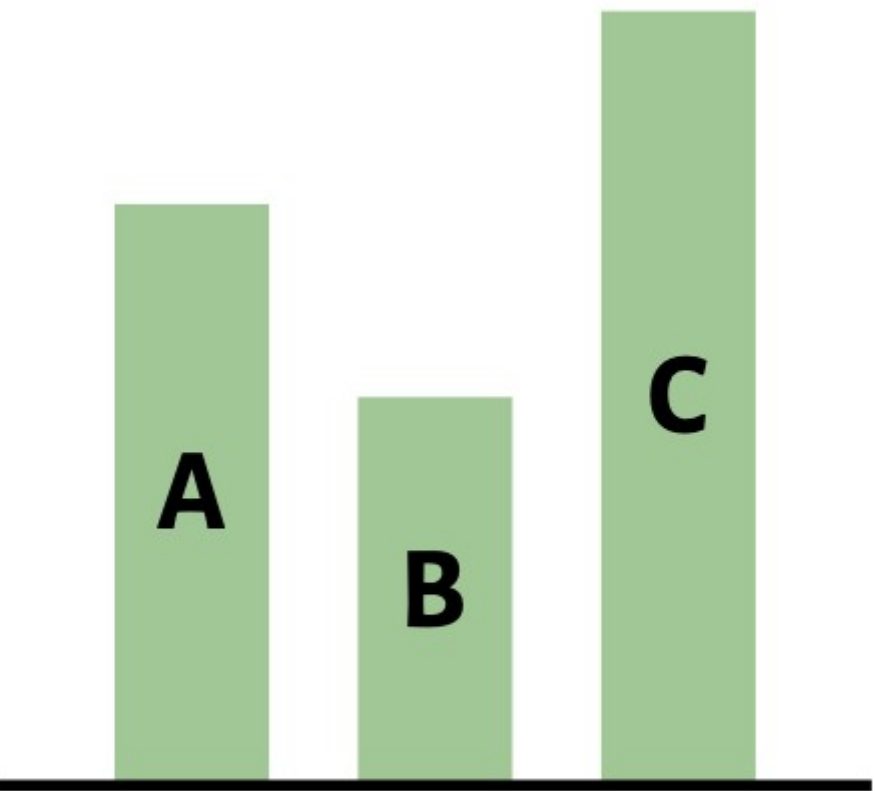
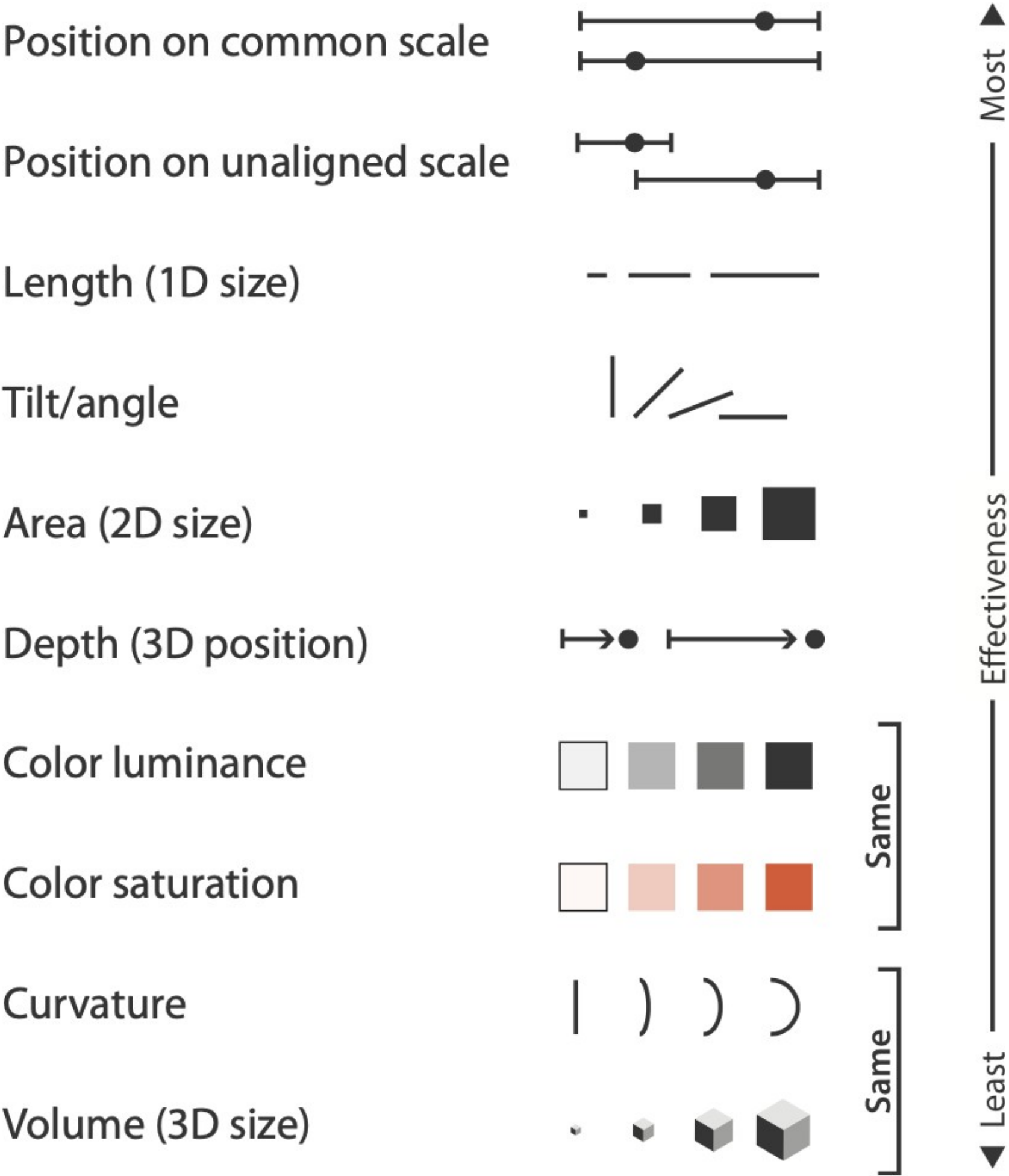
Motion 

Shape 

Tamara Munzner, *Visualization Analysis and Design* (2014).

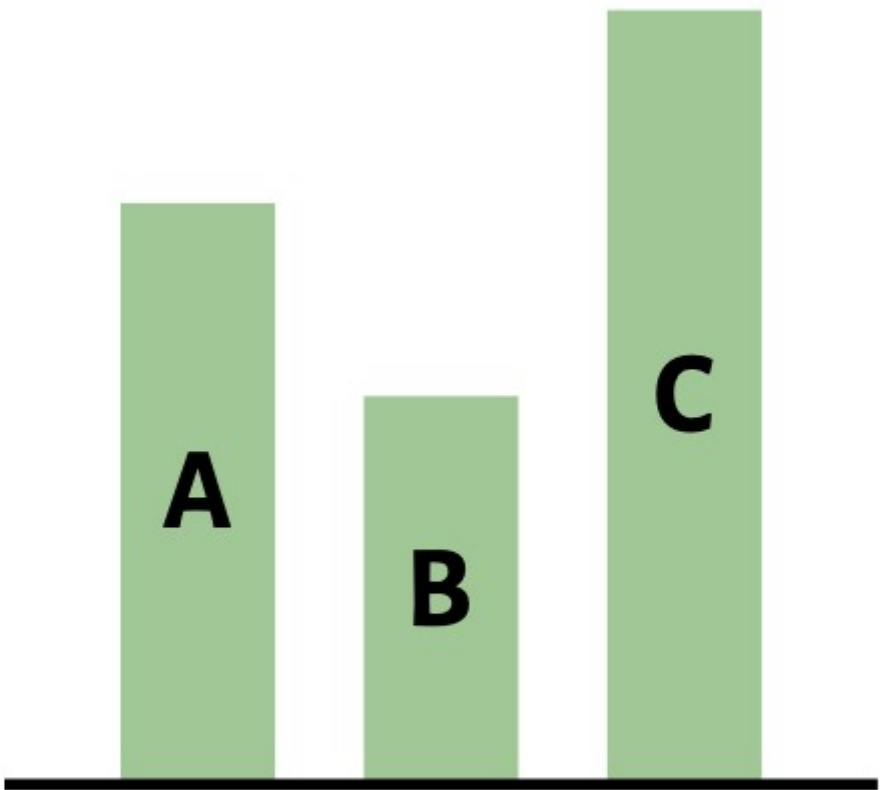
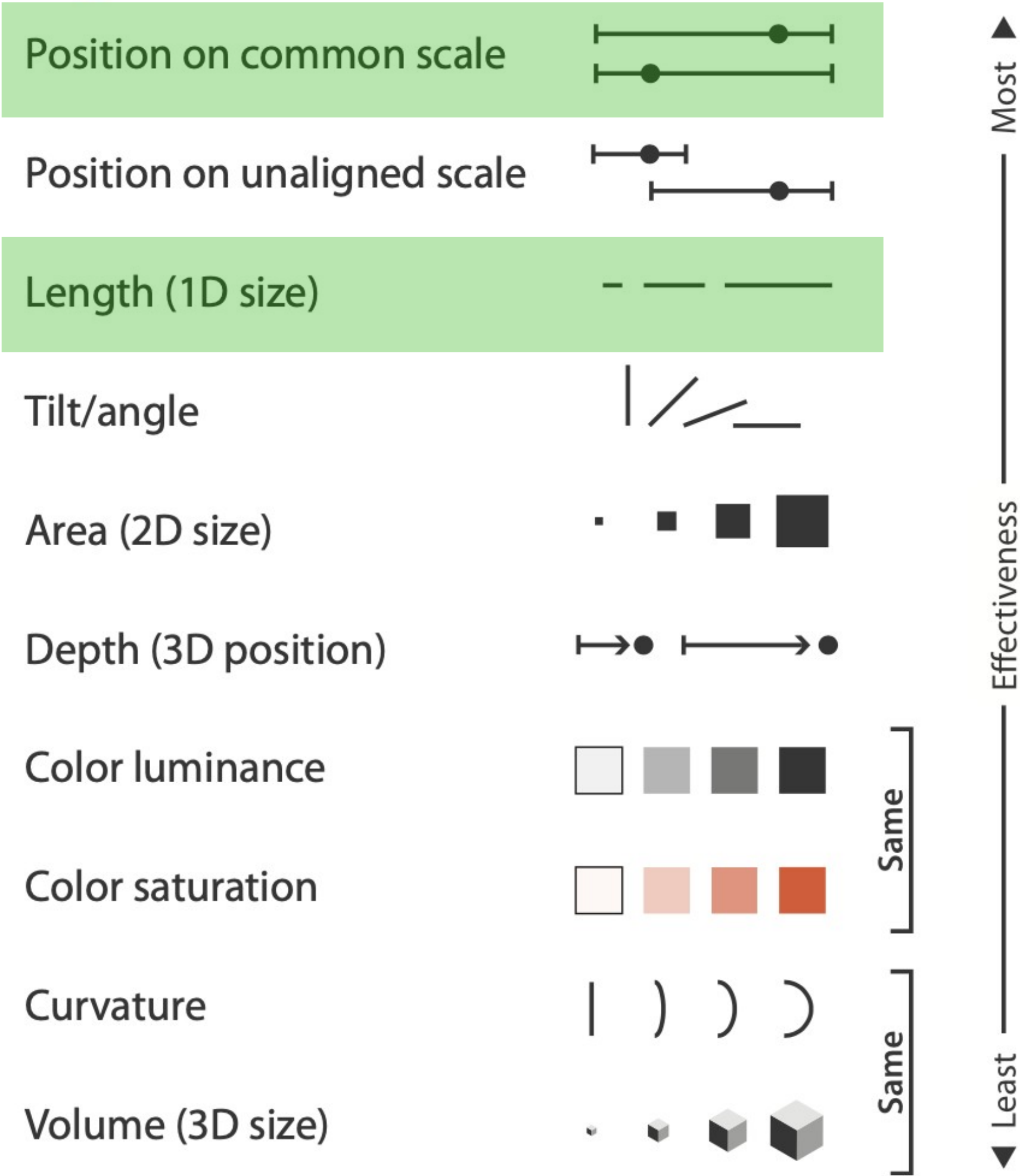
Choose Effective Encodings

➔ **Magnitude Channels:** **O** or **Q** attributes



Choose Effective Encodings

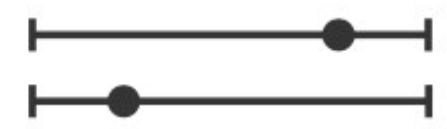
➔ **Magnitude** Channels: **O** or **Q** attributes



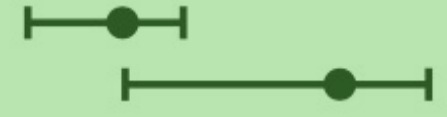
Choose Effective Encodings

➔ **Magnitude Channels:** **O** or **Q** attributes

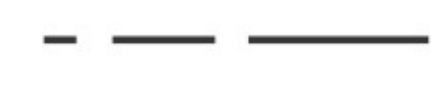
Position on common scale



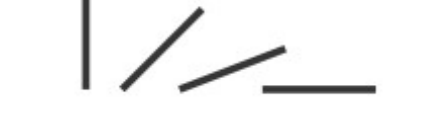
Position on unaligned scale



Length (1D size)



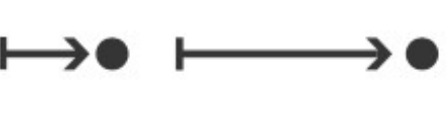
Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



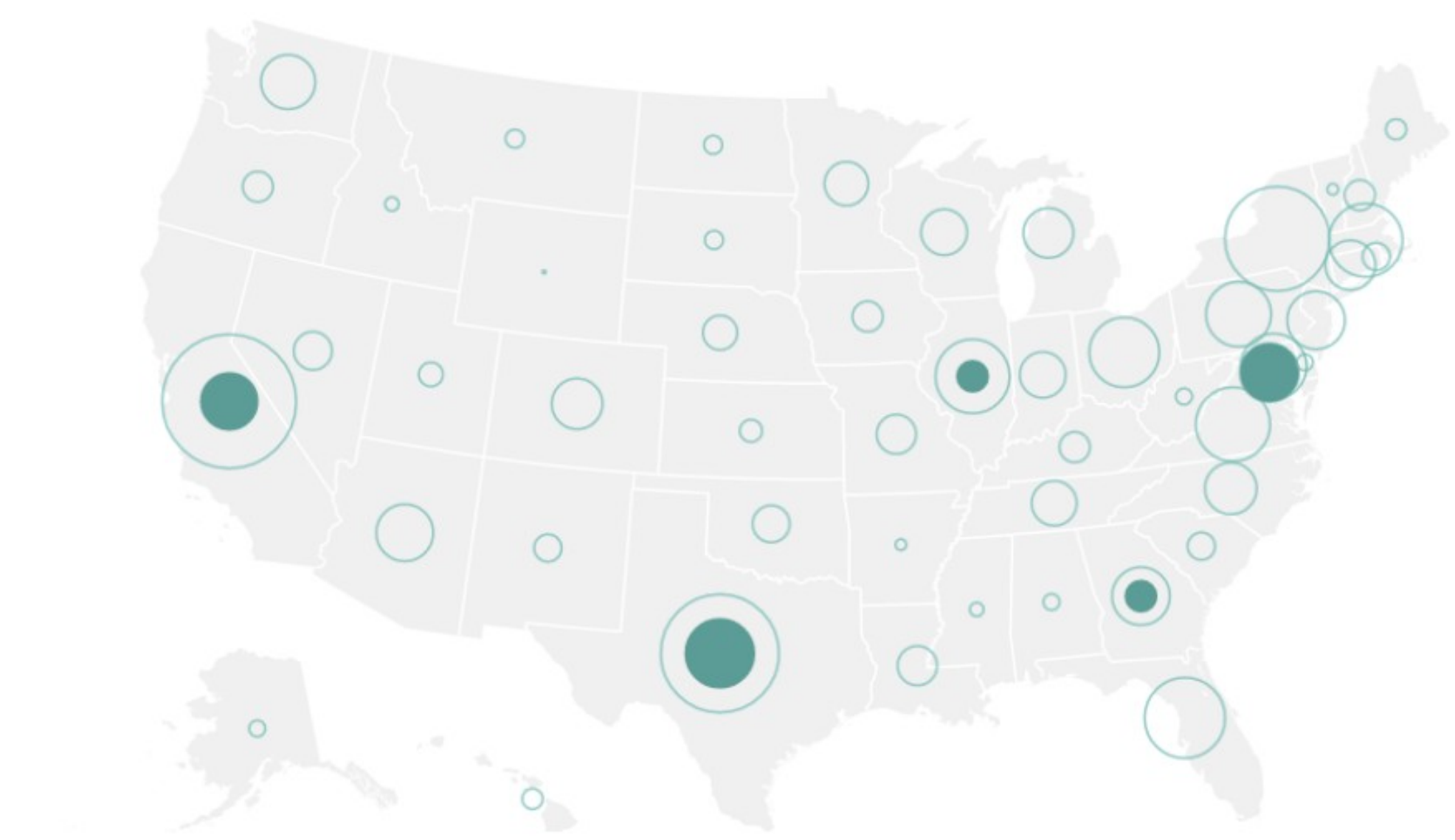
Color saturation



Curvature



Volume (3D size)



Choosing Visual Encodings

Assume k visual encodings and n data attributes. We would like to pick the “best” encoding among a combinatorial set of possibilities of size $(n+1)^k$

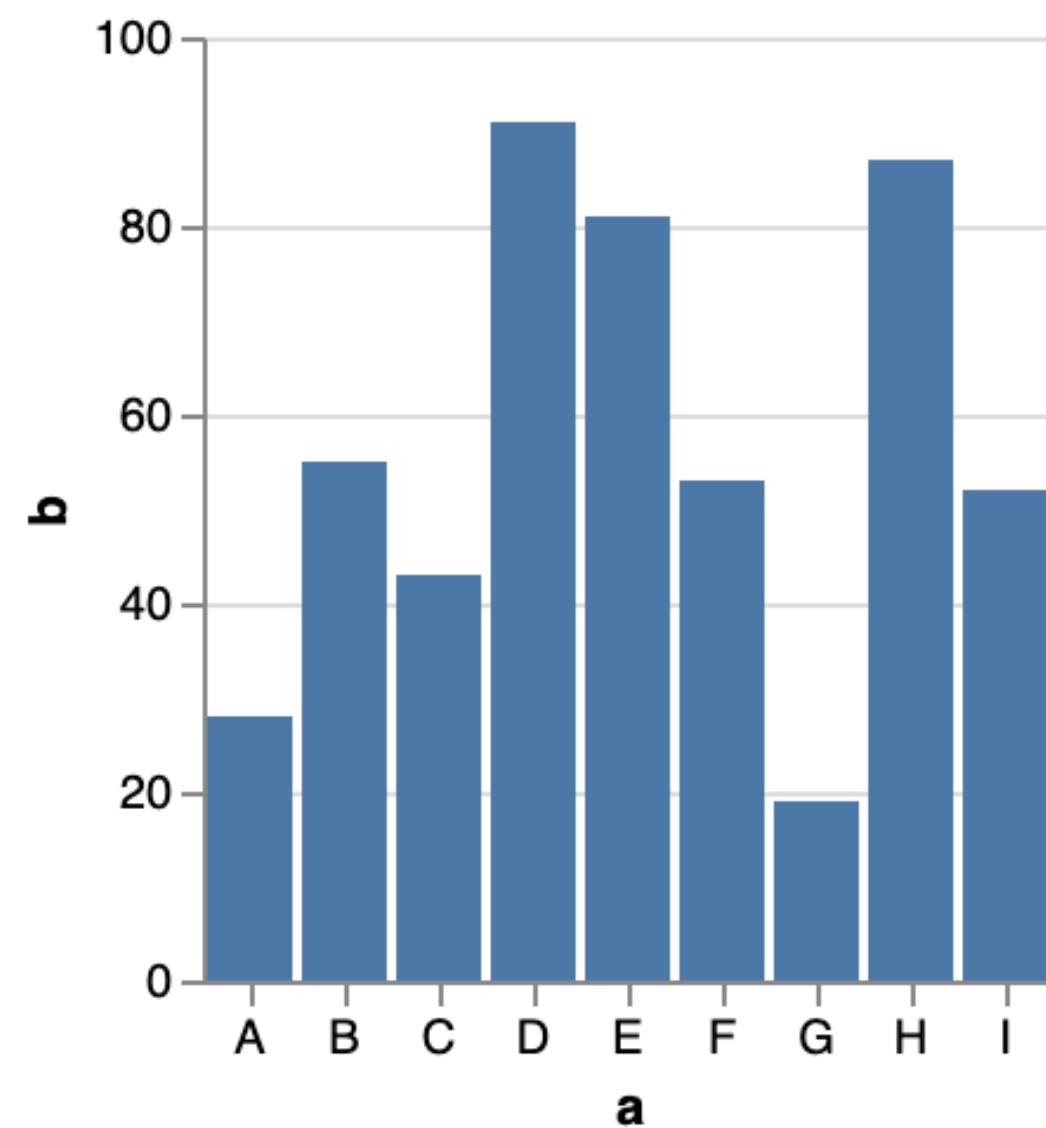
Principle of Consistency

The properties of the image (visual variables) should match the properties of the data.

Principle of Importance Ordering

Encode the most important information in the most effective way.

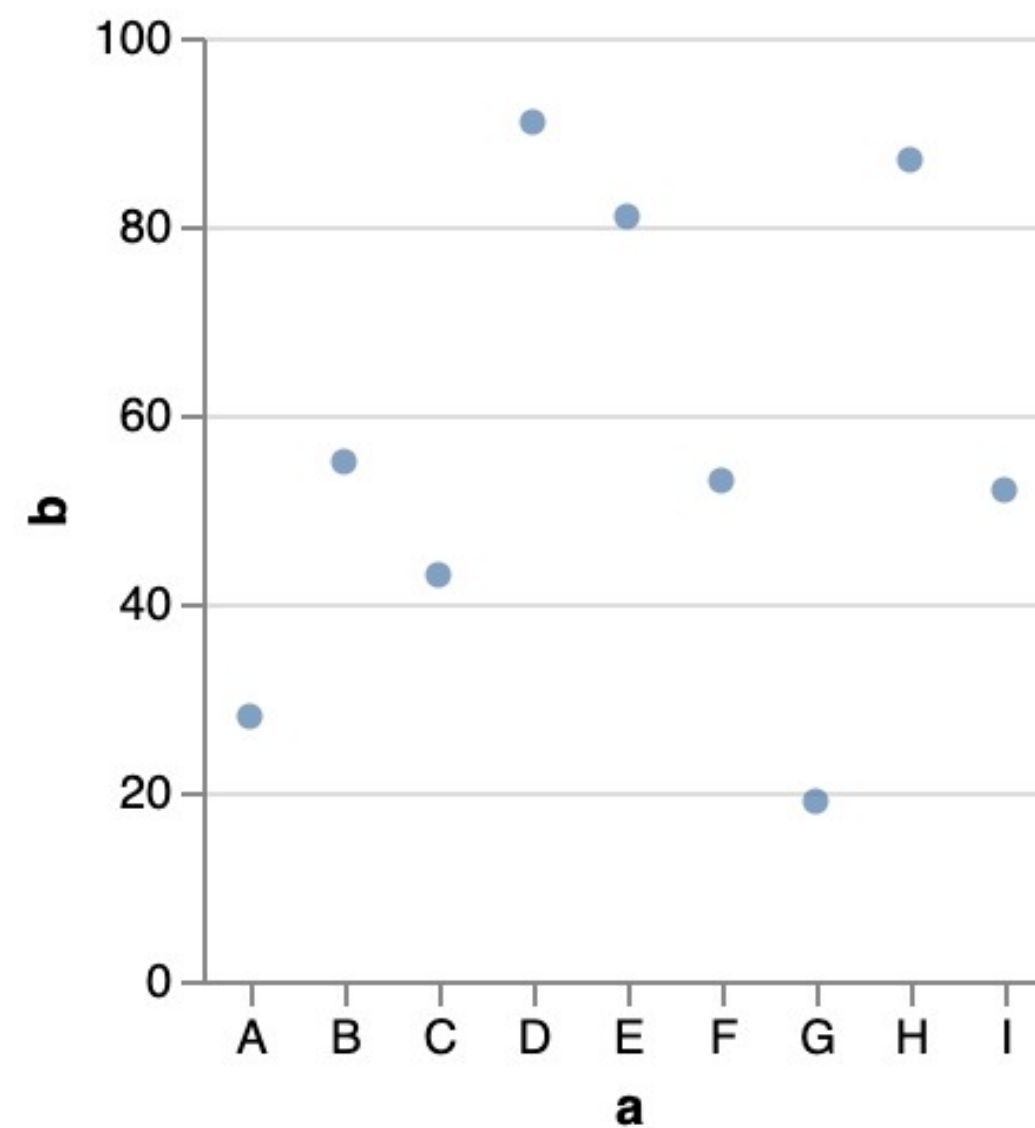
Visual Encoding: 1 Nominal, 1 Quantitative



Mark: Bar

$d_{\text{nominal}} \rightarrow x$

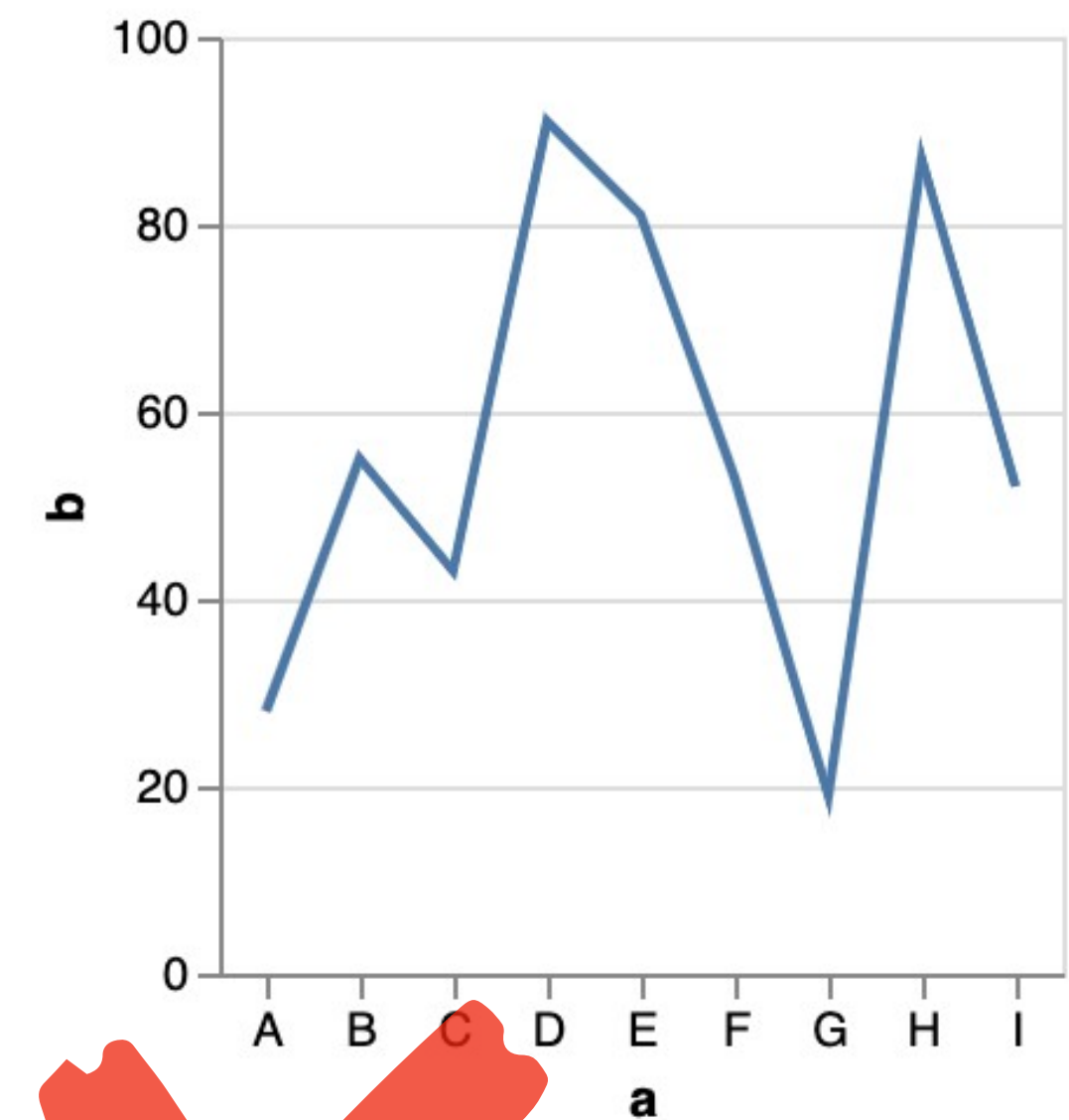
$d_{\text{quantitative}} \rightarrow y$



Mark: Point

$d_{\text{nominal}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$



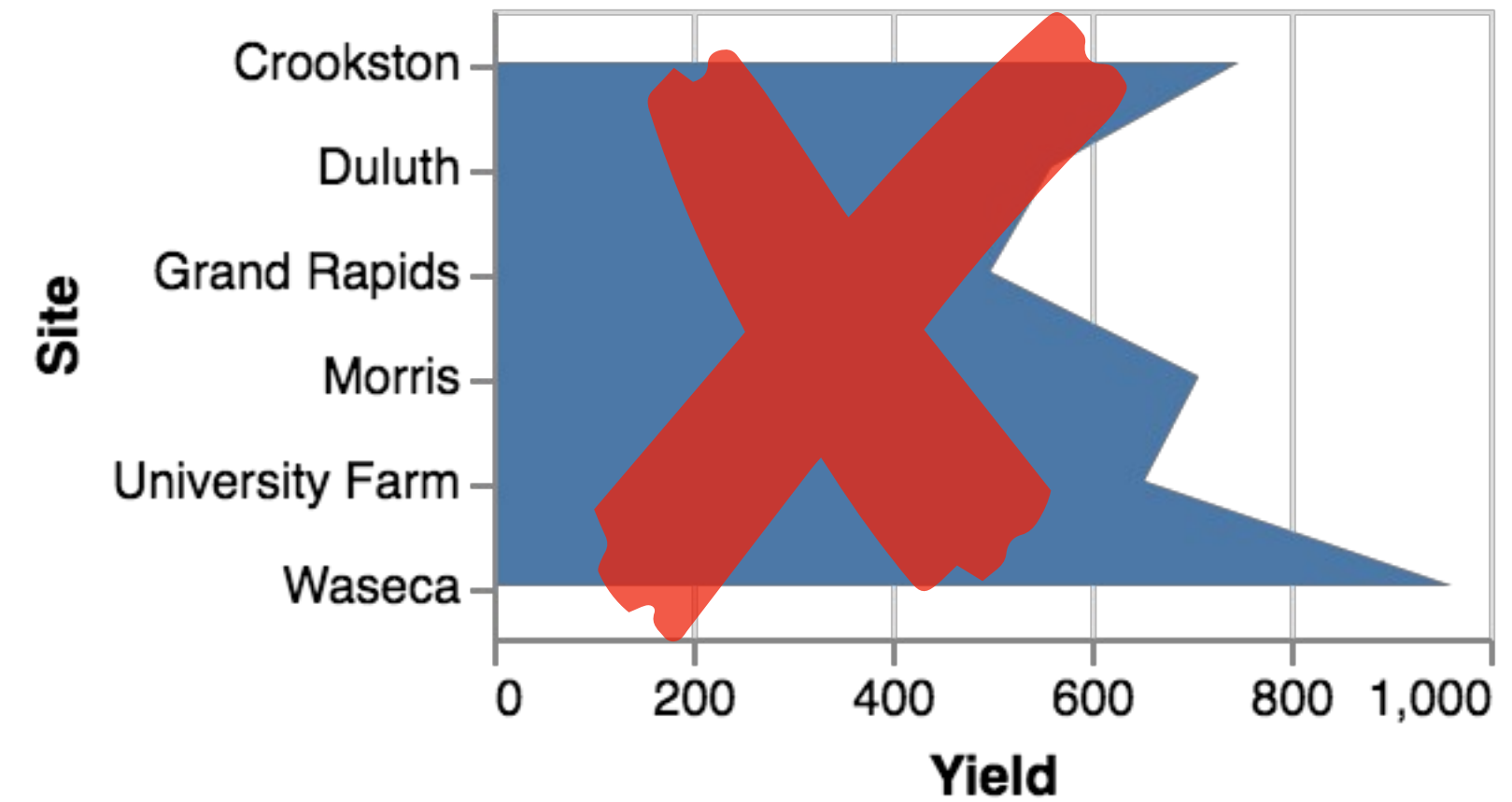
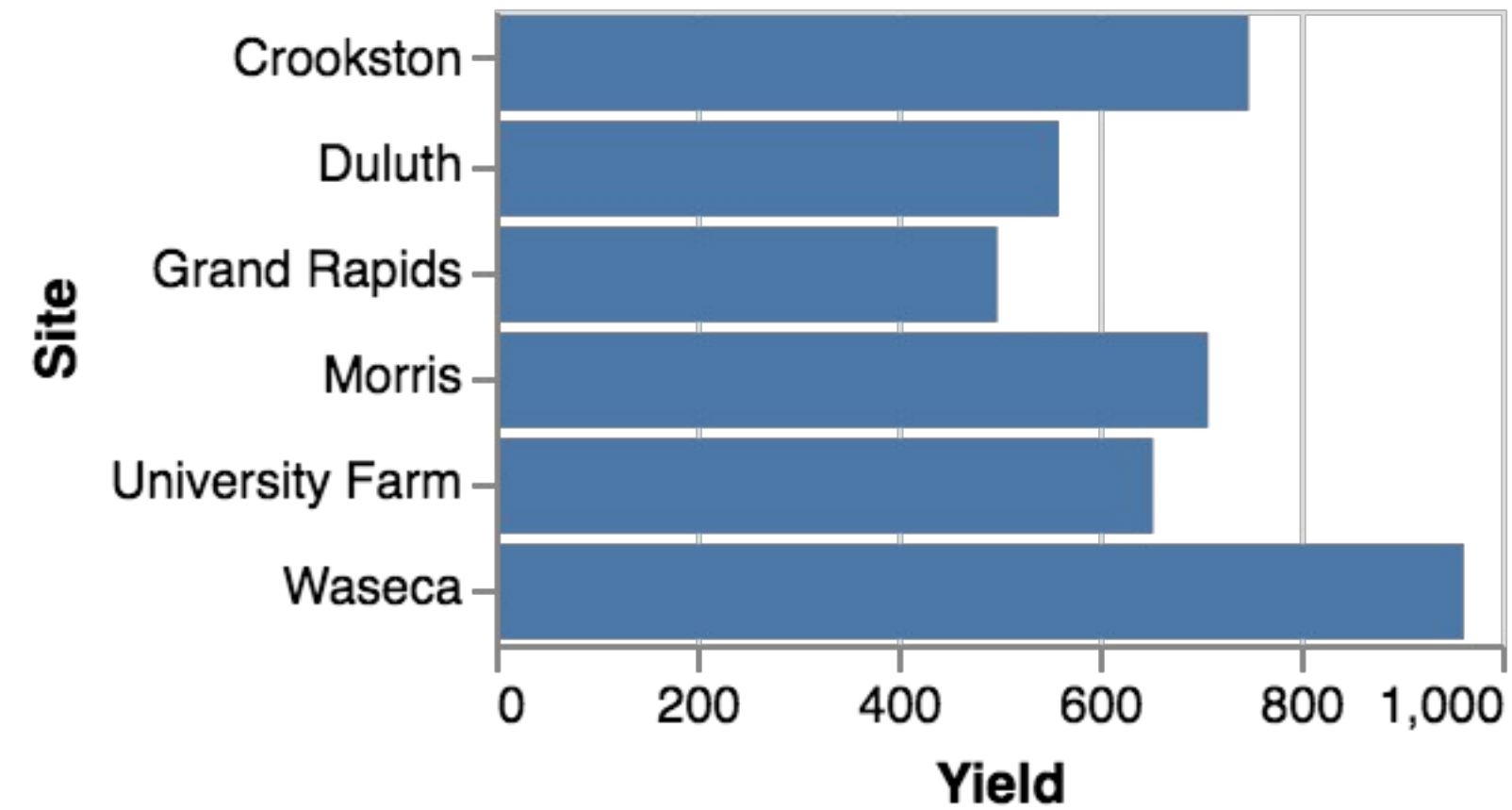
Mark: Line

$d_{\text{nominal}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

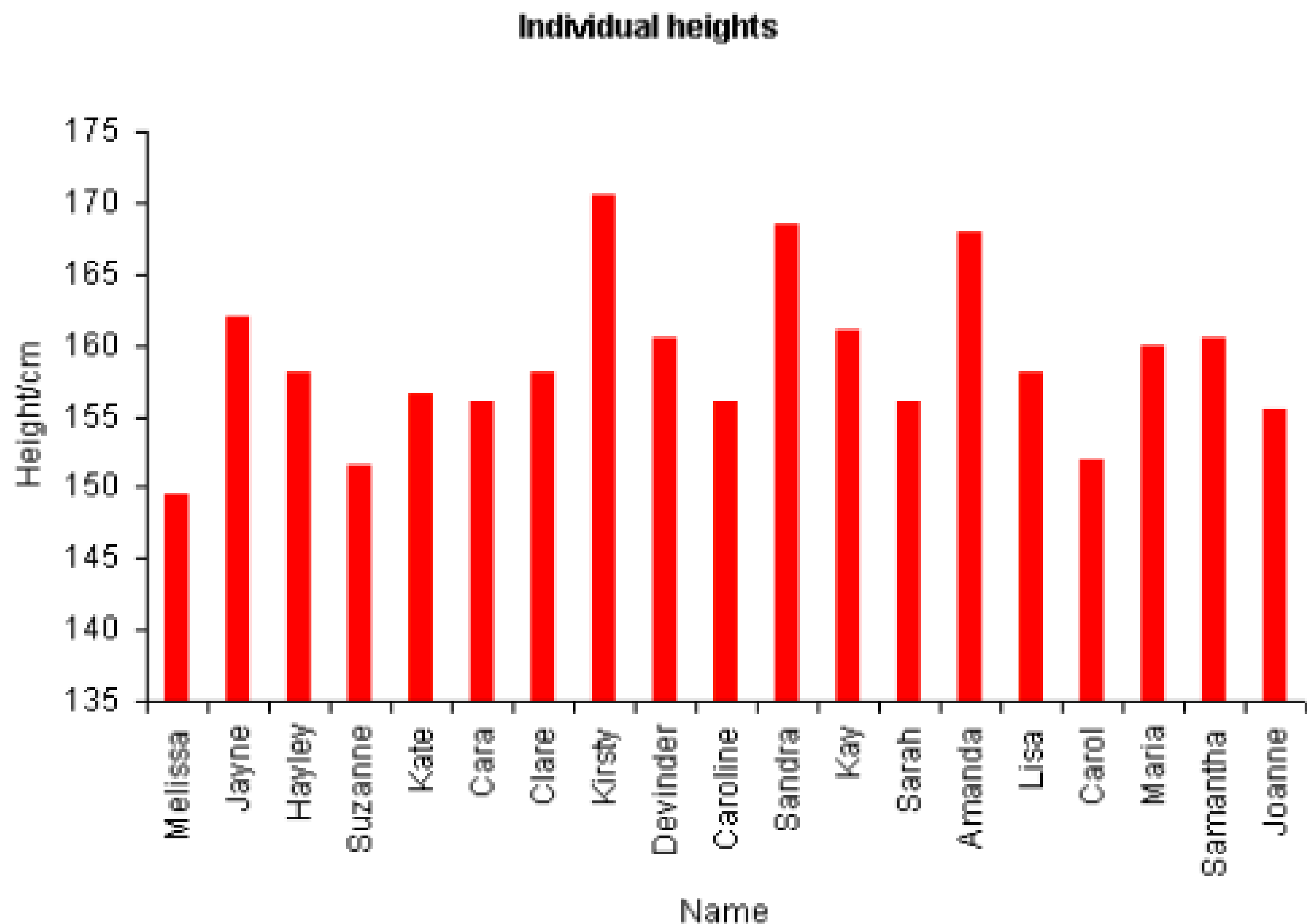
Violates expressiveness: the line mark implies a trend across the categories.

Visual Encoding

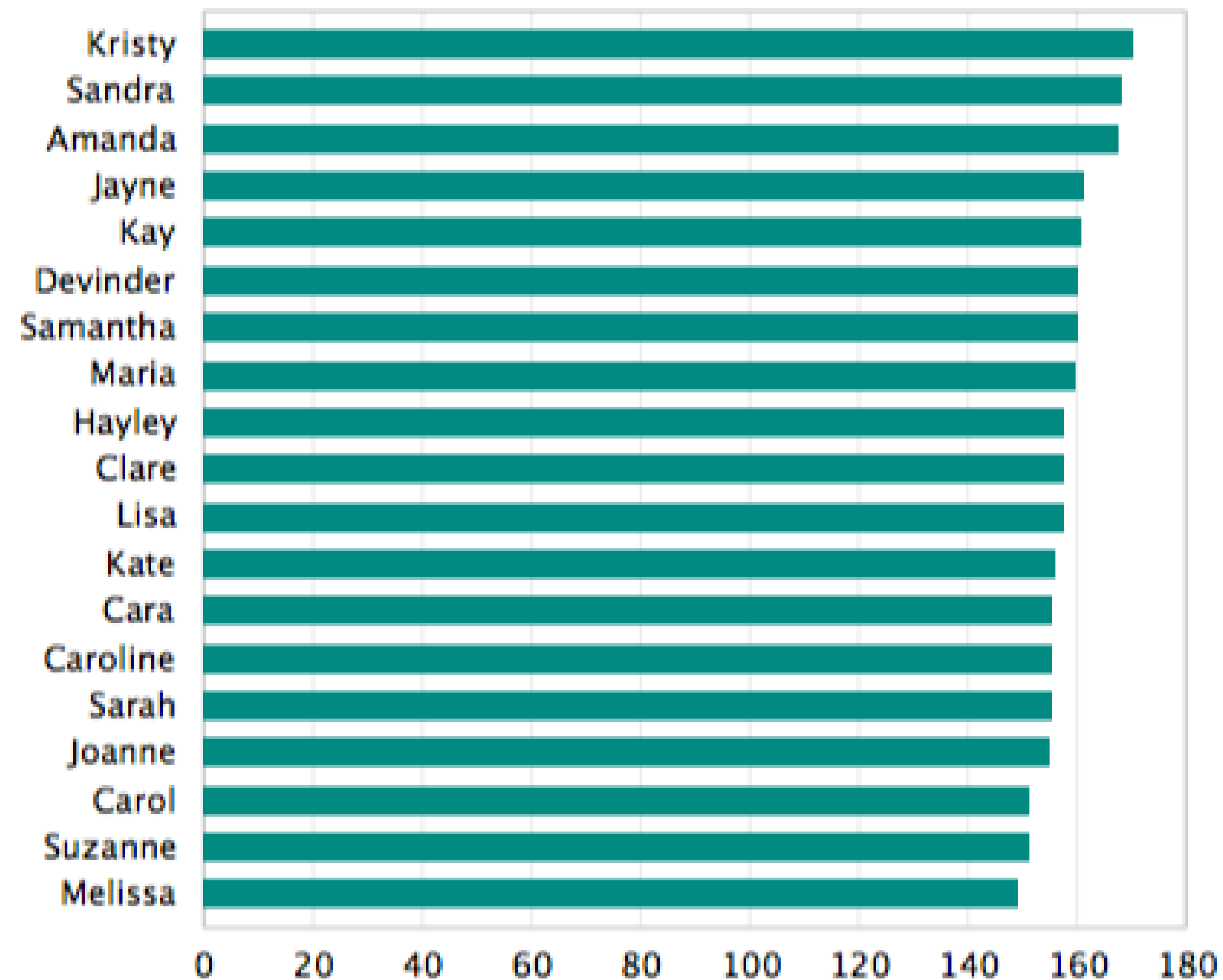
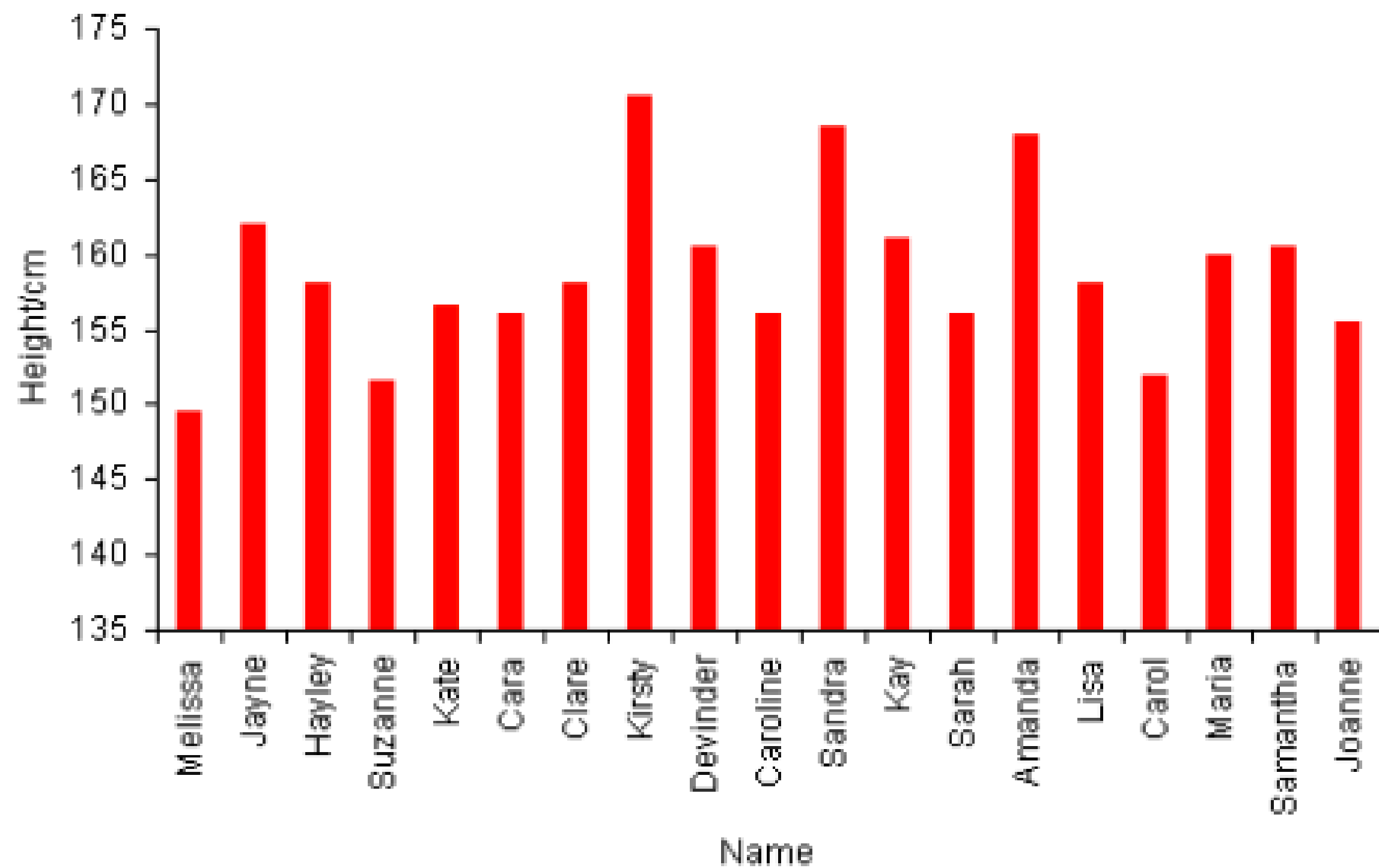


Violates expressiveness:
the area mark implies a trend
across the sites.

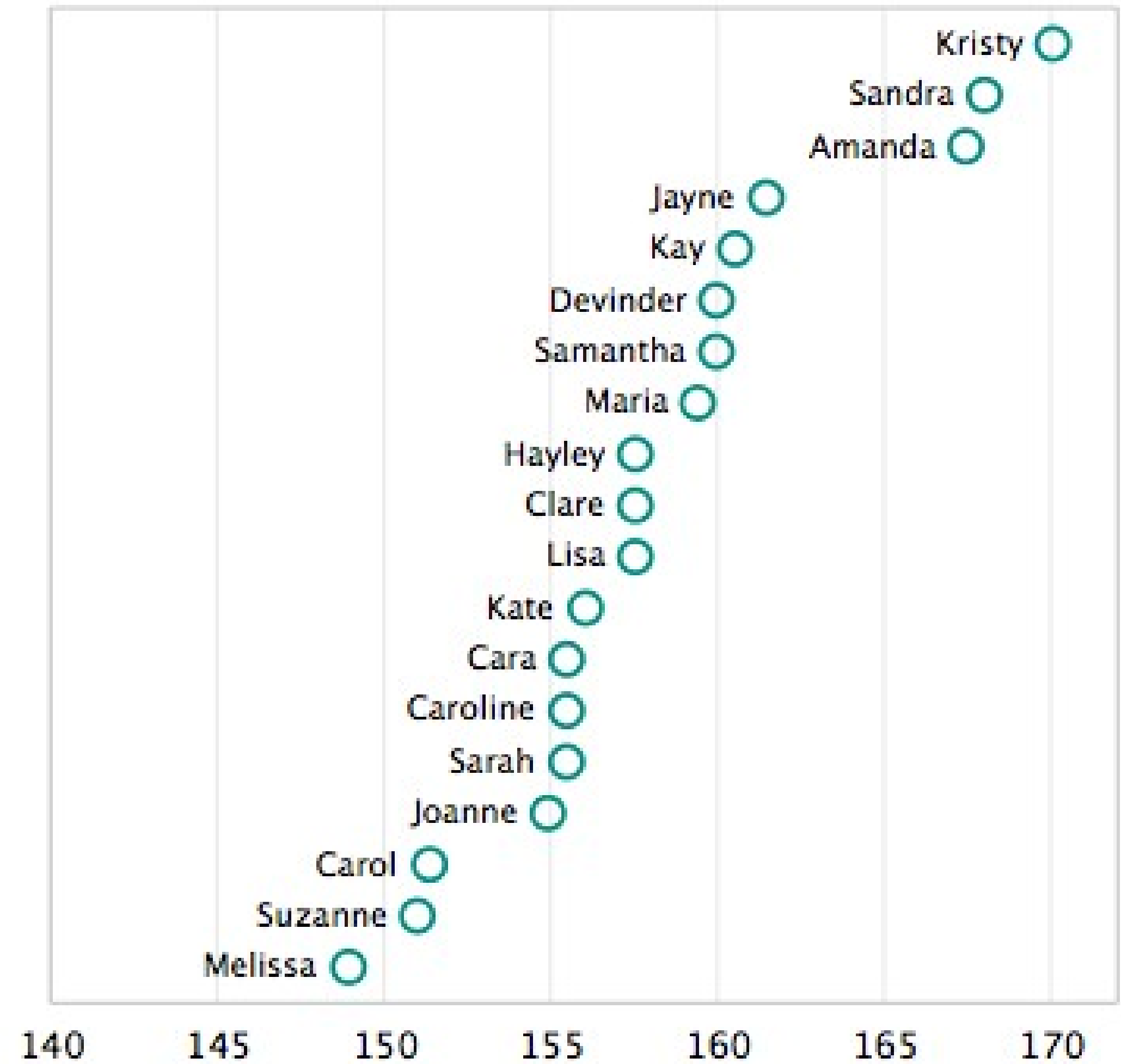
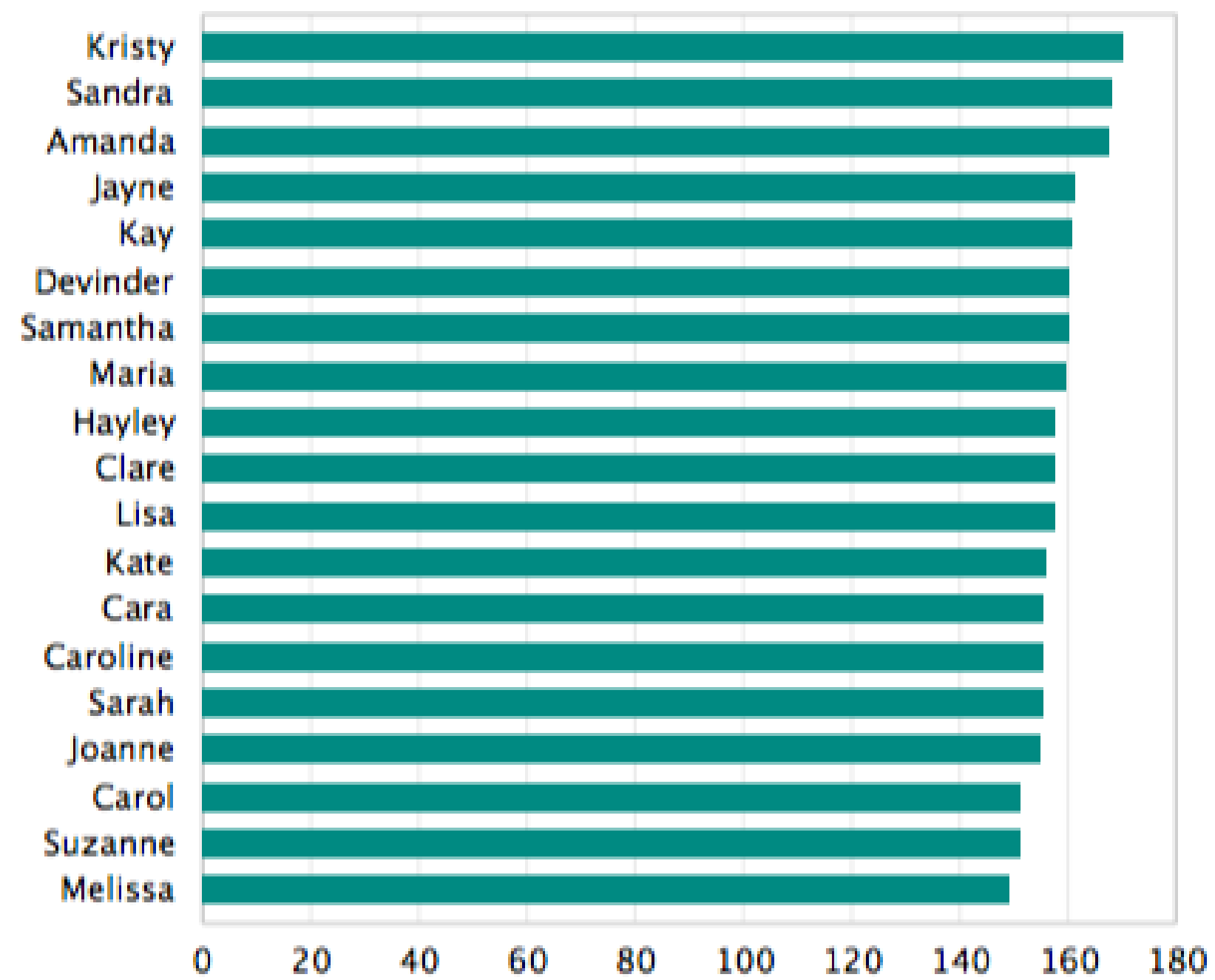
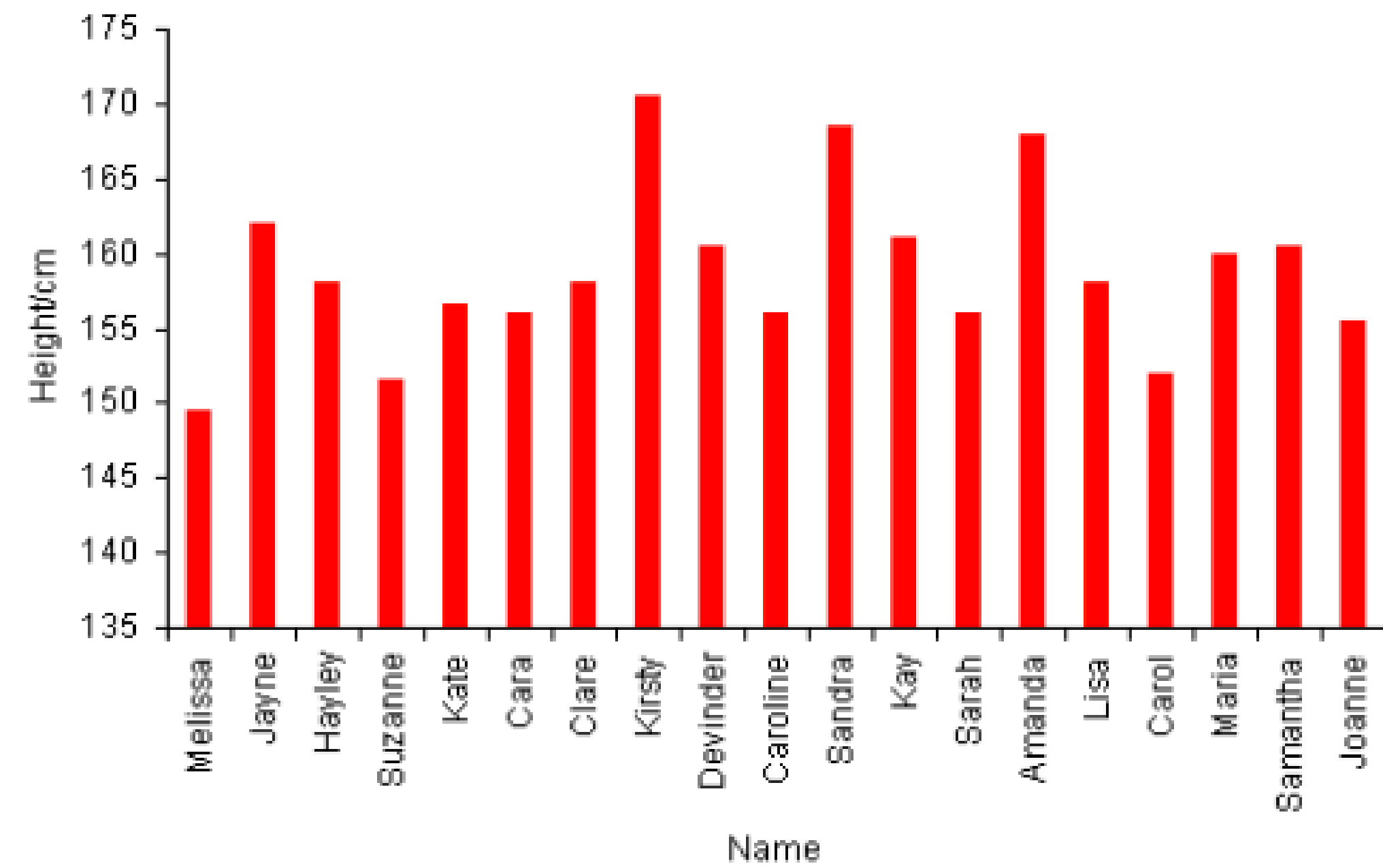
Why might this visualization be inexpressive?



Individual heights



Individual heights



Tufte's Integrity Principles

Show **data variation**, not design variation

Clear, detailed, and thorough **labeling** and **appropriate scales**

Size of the **graphic effect** should be **directly proportional to the numerical quantities** (“lie factor”)

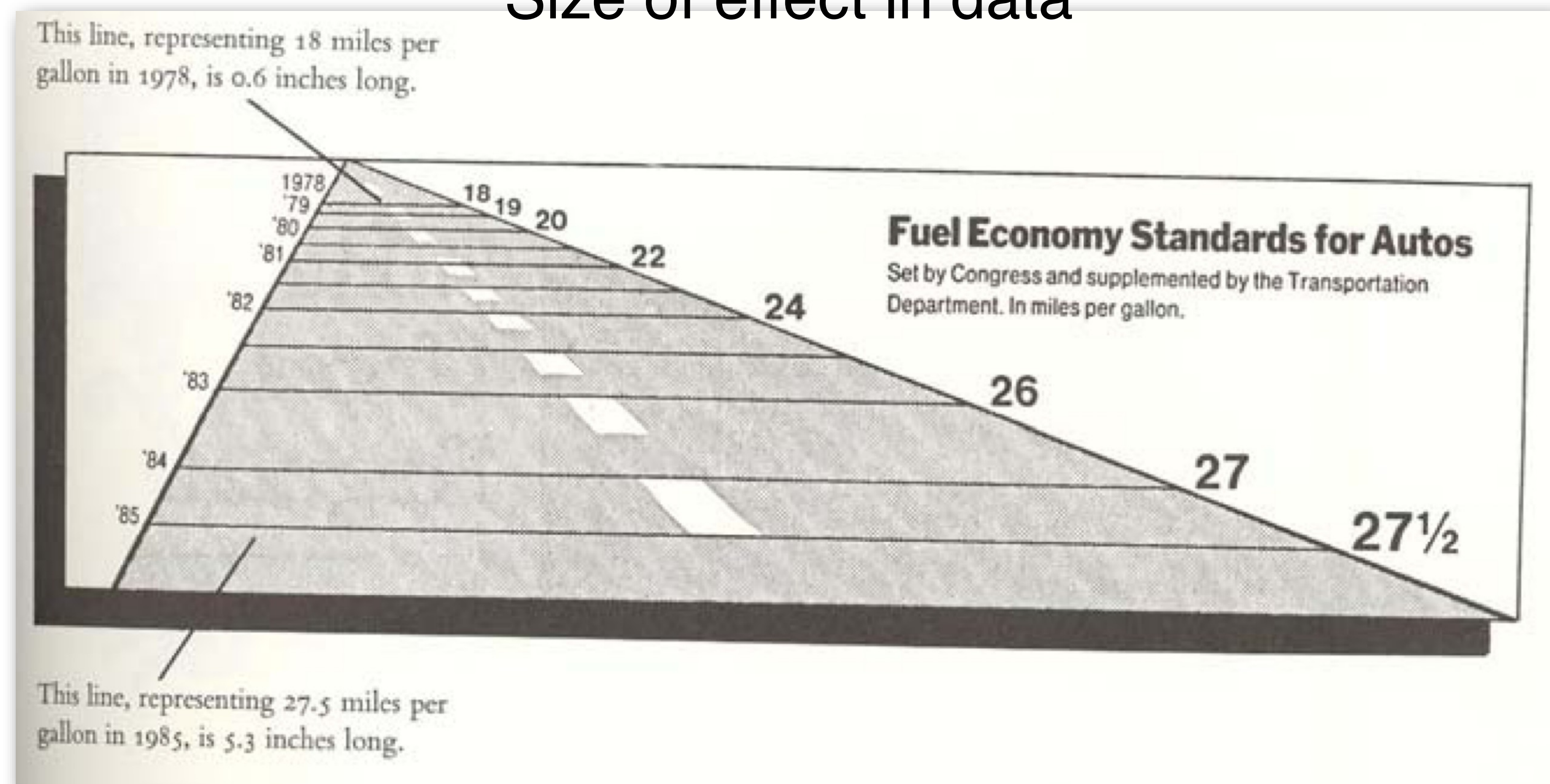
Tufte's "The Lie Factor"

Size of effect shown in graphic

Size of effect in data

The Lie Factor

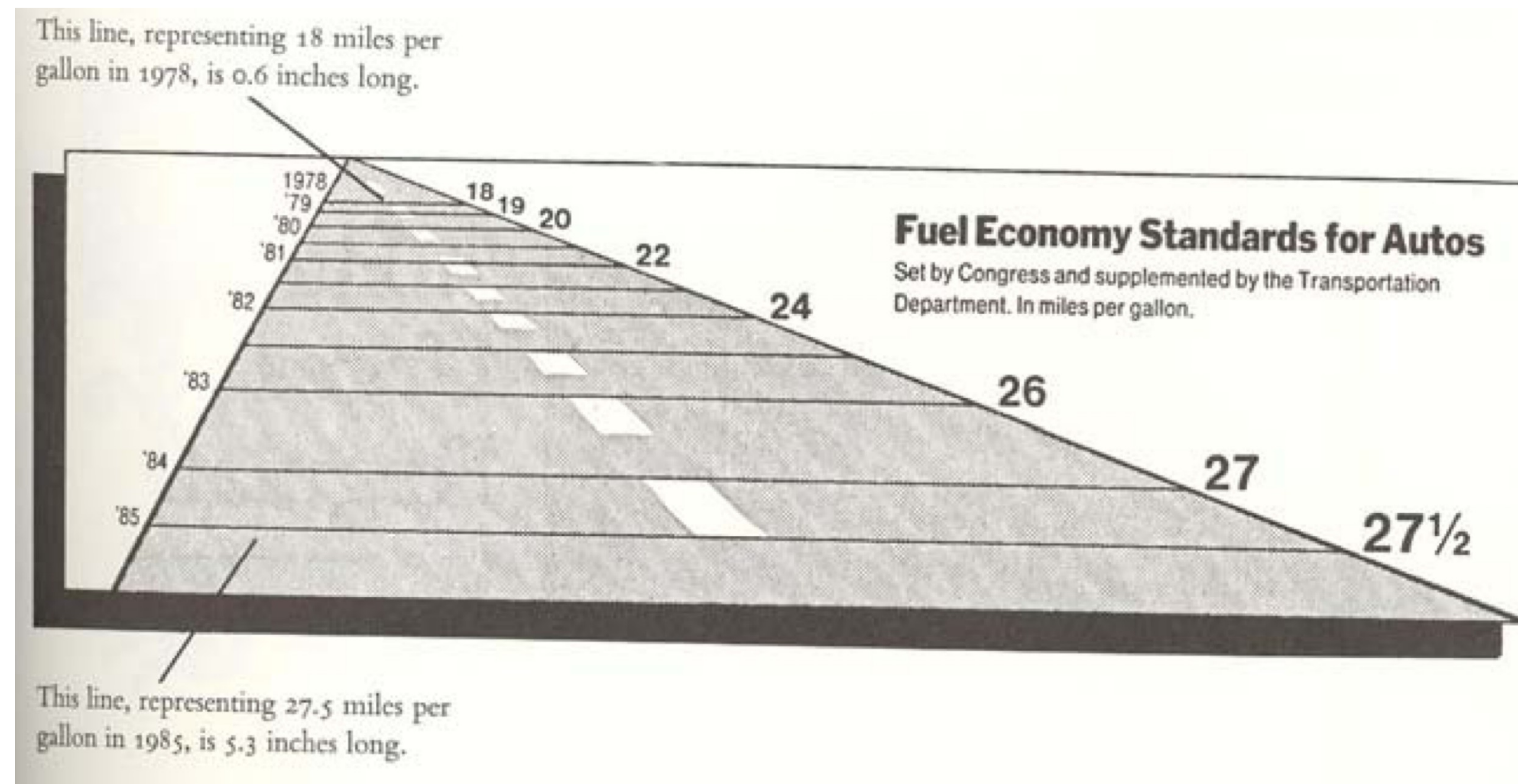
Size of effect shown in graphic
Size of effect in data



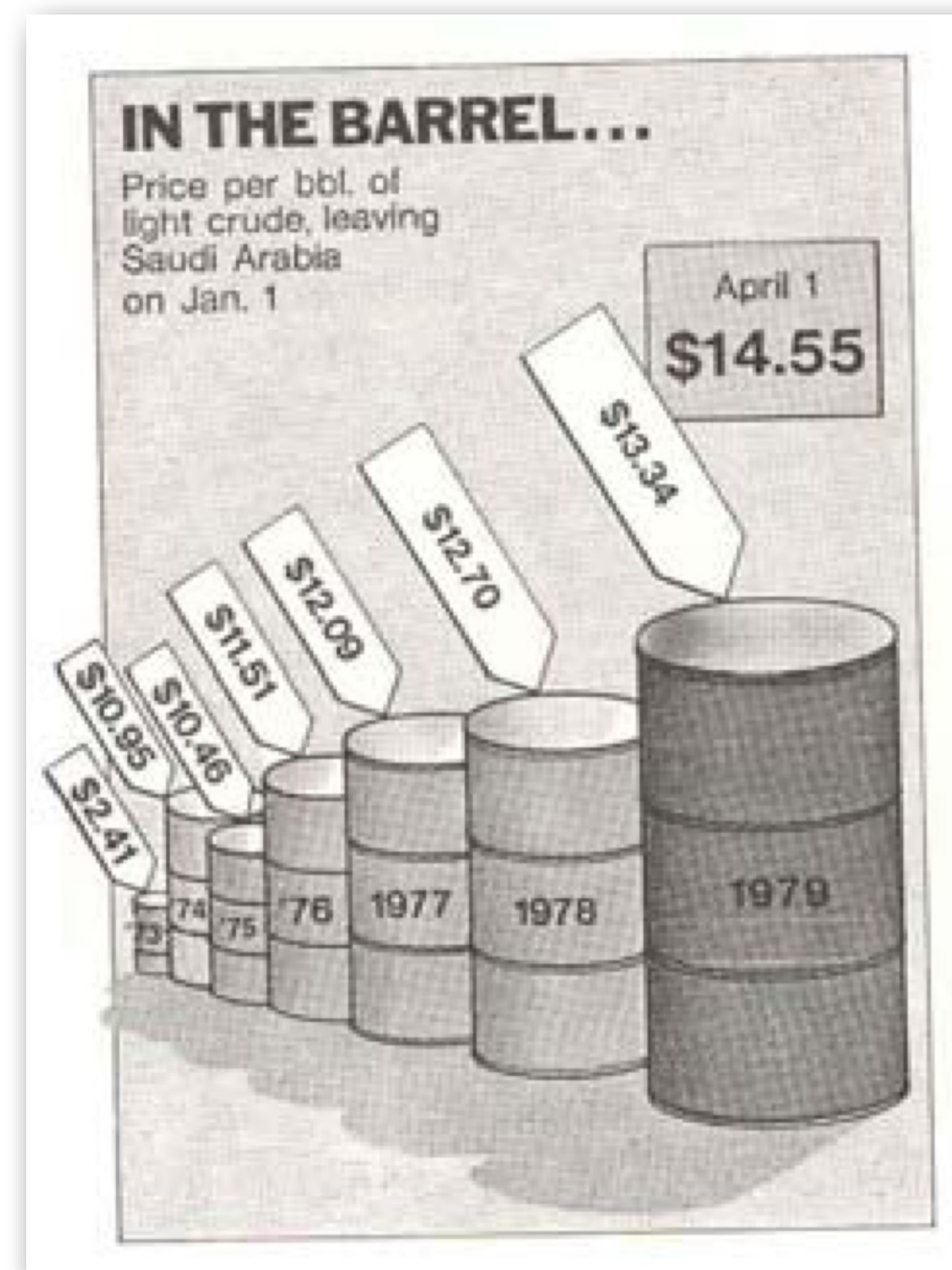
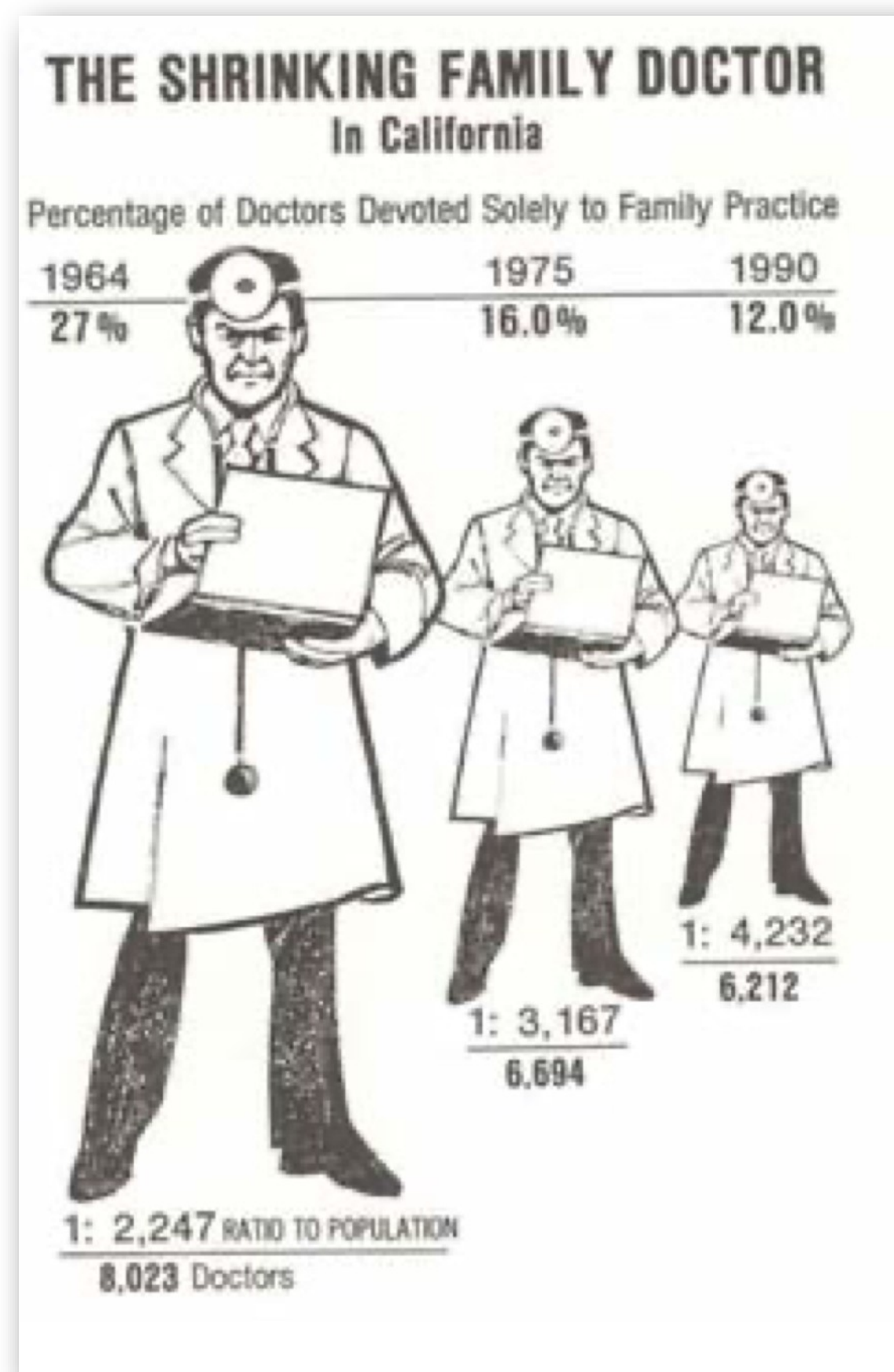
The Lie Factor

$$\frac{5.3 - 0.6}{0.6} / \frac{27.5 - 18}{18} = 14.8$$

(Size of effect in graphic)/(size of effect in data)



The Lie Factor



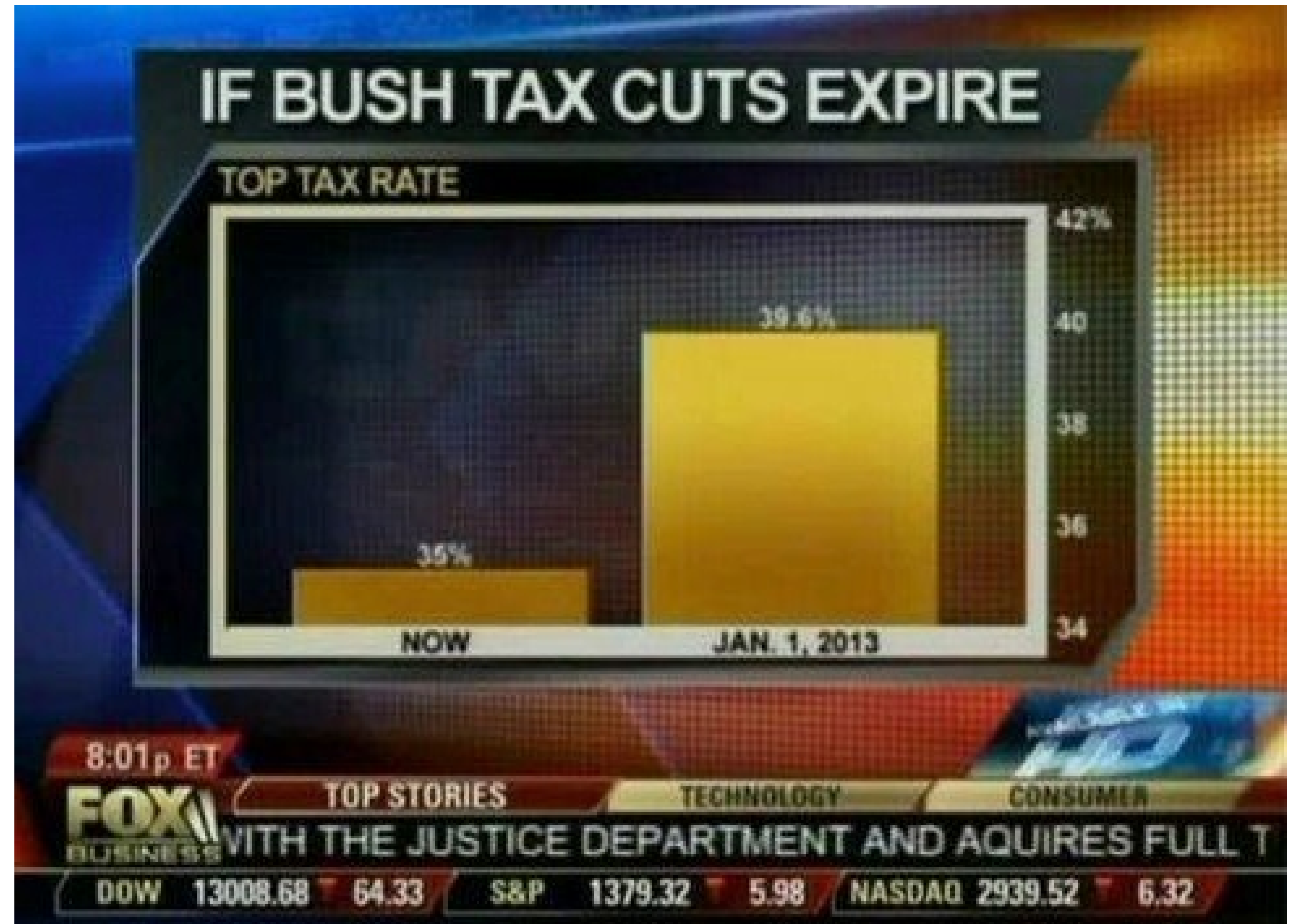
Lie Factor - Graphical Integrity

Magnitude in data
must correspond to
magnitude of mark

Effect in Data: factor 1.14

Effect in Graphic: factor 5

Lie Factor: $5/1.14 = 4.38$



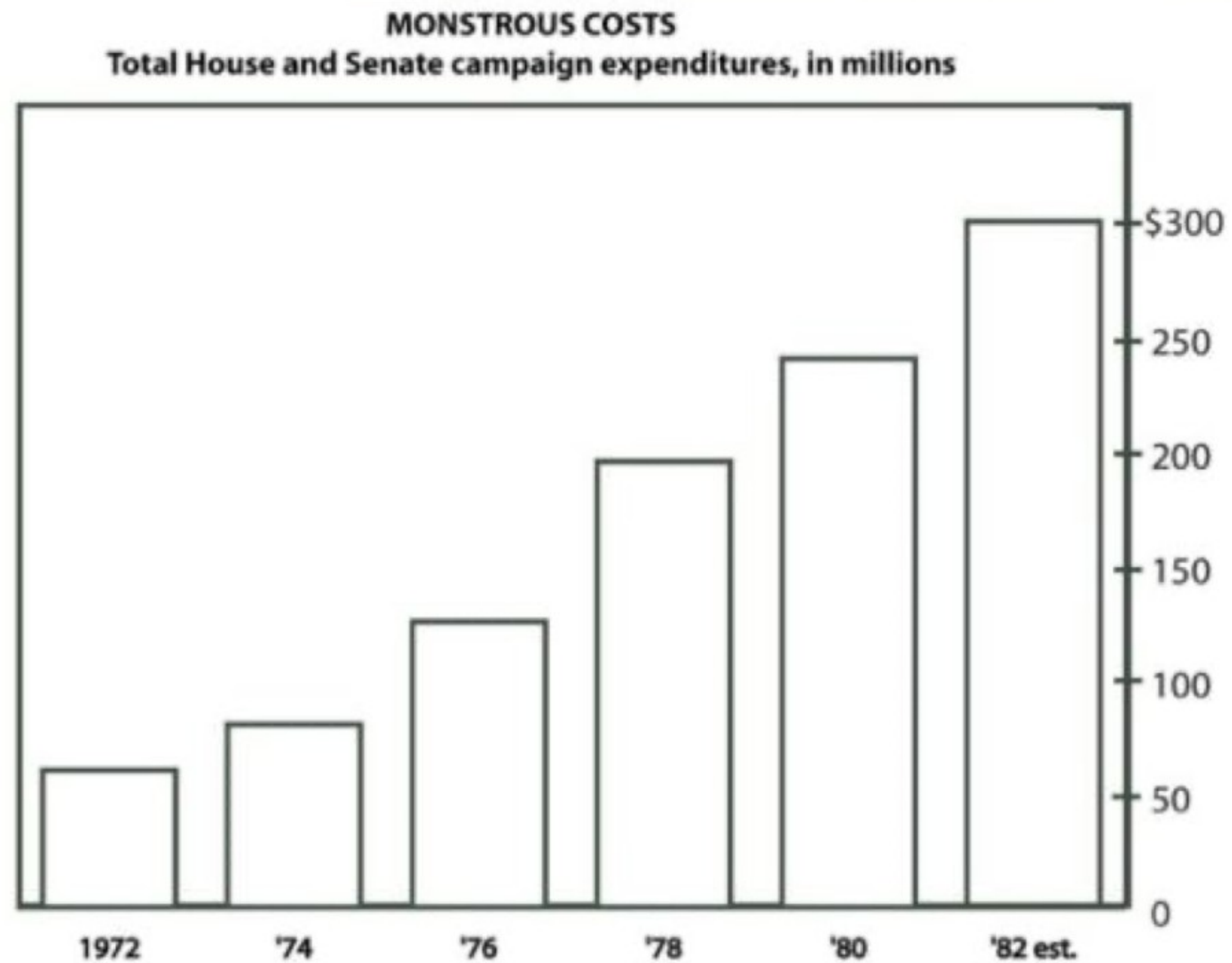
Guiding Principles: steps for building a chart

1. First, **identify attribute types** in the data and the **goal** of the graphic.
2. Consider **effectiveness ranking** for attribute type to determine **channel** (identity or magnitude)
 - Encode most important information with most effective channel first
3. Choose **scales** with appropriate **range** for optimal aspect ratio and **scale type** to change the question being addressed
 - Consider if you should **start at 0 or something else**
 - Use well-marked **scale breaks** sparingly
4. Use **hue** for unordered (**nominal**) attributes and sequential or diverging colormaps (hue + **luminance** or **saturation**) for **ordered** attributes
5. Consider the **Lie Factor** of your visualization → should be close to 1

**Now that we know
the rules...**

**we can break the
rules.**

Effectiveness & expressiveness vs. Memorability



Which of the previous
charts was more
effective?

Which was more
memorable?

accuracy in describing the embellished charts was no worse than for
plain charts, and their *recall* was significantly better